The biosecurity collective-risk social dilemma: Simulating the prevention of a rapidly spreading pest

Abstract

Protection against pest invasion is a collective-risk social dilemma. The overall protection in a region of actors depends on the biosecuring actions they each adopt. If too few take protective measures, then the entire region is at risk, since if a pest colonises one property it can quickly spread to neighbouring properties, causing catastrophic economic losses for all. Here we introduce an economic game—the biosecurity collective-risk social dilemma—for simulating the problem of collective pest management. In the game, four-player groups interact over a series of rounds. In each round, players must decide whether to pay a private cost to take biosecurity action to protect a value at risk. The game is based on a one-shot game-theoretic model of collective pest management developed by Hennessy (2008). According to the model, biosecurity decisions depend on the expected net benefit of protection, which varies with the risk of pest entry and the extent to which others also take protective action. We additionally hypothesised that the opportunity for actors to communicate should increase engagement in risk-reducing action. We tested these predictions in an experiment where we manipulated the risk of an outbreak occurring and whether group members could communicate on certain rounds. In the experiment, both higher background risk and communication increased biosecuring actions. Communication was effective because it increased trust, provided opportunities for conditional cooperation, and facilitated strategic coordination. The results suggest additional effort should be directed to improving risk communication and ensuring biosecurity schemes include a strong element of communication.

Keywords: biosecurity · communication · collective-risk social dilemma · cooperation · conditional cooperation · fruit fly · risk

1 Introduction

Globalisation, by increasing trade and travel, has increased invasions of exotic agricultural pests and diseases and ecosystems threats, such as Myrtle Rust (Carnegie, 2015), Khapra beetle (Athanassiou et al., 2019), and red imported fire ant (Wylie & Janssen-May, 2017). Alongside exotic pests, farmers worldwide also face evolving endemic threats. The pesticides at their disposal to address these problems are becoming more restricted and many are no longer available (Dominiak & Ekman, 2013). Simultaneously, export market sensitivity to exotic pest risks and pesticide residues is increasing.

There is mounting evidence that the standard response, where a farmer takes an individual decision to apply a pesticide, is no longer valid (Bagavathiannan et al., 2019; Garcia-Figuera et al., 2021). A farmer's decision to control a pest cannot be seen in isolation as there are spillover effects for neighbouring farmers and regions. A farmer who chooses to do nothing about a pest—because it is individually rational—imposes costs on their neighbours. Similarly, the efforts of proactive farmers may be dissipated if their neighbours shirk on pest management. For some pests, traditional pest control solutions where individual growers respond on a farm-by-farm basis are likely to be short-lived, expensive, or completely ineffective. The solution is cooperative integrated-pest management and area-wide management where coordinated efforts reduce pest populations at the regional level (De Lima et al., 2025; Florec et al., 2013; Tam et al., 2023; Tapsuwan et al., 2020).

No pest exemplifies the need for cooperation between farmers more than the global struggle against endemic and exotic fruit flies—fruit flies devastate crops and close export markets, generating losses amongst large groups of farmers (Florec et al., 2013; Sultana et al., 2020; Tam et al., 2023). Export market closure can be triggered by a relatively minor fruit fly outbreak. Even short-term export market closures can be very costly for producers of highly perishable products, as it can lead to a complete loss of the crop revenue.

1.1 Biosecurity as a public good

Biosecurity is a collective action problem or social dilemma involving a conflict between individual and collective interests (Bagavathiannan et al., 2019; Donaldson, 2008; Garcia-Figuera et al., 2021; Hennessy, 2008; Maru et al., 2025). A social dilemma arises when a group of individuals must cooperate to achieve a shared goal. The socially optimal outcome arises when all individuals cooperate but because each person can benefit from the cooperation of others, there exists an incentive to defect. In the case of biosecurity, an individual who chooses to biosecure benefits their neighbours by reducing the risk of a pest outbreak that could spread to surrounding properties. Conversely, an individual who does not biosecure imposes costs on their neighbours by increasing this risk. Accordingly, an individual's protection from pest invasion depends on both their actions and those of others in the region. The risk they face is an *endogenous* collective risk—rather than an individual risk—that can only be mitigated through cooperation.

Of the various types of social dilemmas, biosecurity is a public goods dilemma, wherein the public good is protection against exotic pest invasion and its provision requires cooperation between individuals who must adopt biosecuring actions that incur private costs. The dilemma arises because an individual's ability to benefit from low pest-invasion risk does not diminish the ability of others in the affected region to enjoy those benefits (non-rivalry), and no individual can be excluded from obtaining the benefits (non-excludable). These properties of the public good render it vulnerable to free riding. That is, a self-interested individual has an incentive not to biosecure, if they think that a majority of their neighbours will biosecure. In this way, they may obtain the benefits of the public good, without paying the costs of its provision. However, if each individual thinks this way, then the public good will never be supplied.

Originating in experimental economics, economic games have become an important tool for studying human behaviour in social interactions (Pisor et al., 2020; Thielmann et al., 2021), including social dilemmas. They are abstract formulations of real world

cooperation scenarios and use real outcomes in the form of monetary incentives to motivate participants to act according to their true preferences. Economic games are described and analysed using standard game theory in economics, which, like much economic modelling, makes the simplifying assumptions that individuals are fully rational and perfectly self interested. They are among the most important methods for studying cooperation (Gächter & Herrmann, 2006). Although observational studies provide valuable insights into behaviour by studying participants in their natural environments, in the field many variables are simultaneously operative. The inability to manipulate independent variables directly, whilst controlling for confounding variables, renders it difficult to establish cause-and-effect relationships. Economic games, by comparison, take place in controlled experimental settings to help isolate causal influences on behaviour. Such games have become a key method for testing and modifying economic theories whose strong normative predictions are frequently at variance with actual behaviour (Camerer, 2003; Chaudhuri, 2011; Fehr & Schmidt, 2006; Ledyard, 1995), identifying stylised facts about human cooperation, and uncovering mechanisms by which to improve human cooperative behaviour.

The dominant economic game for studying cooperation is the deterministic public goods game, which simulates the social dilemma underlying public good provision (for reviews, see Chaudhuri, 2011; Ledyard, 1995; Zelmer, 2003). In a typical game, there are groups of four players. In each of several rounds, players are endowed with a sum of money and must decide how much to contribute to a public account without communicating. Any money contributed gets multiplied by some factor before being distributed equally amongst the players, irrespective of whether they contributed to the public account. The socially optimal outcome results when all players contribute all of their money to the public account. However, standard economic theory predicts that a self interested player that is only motivated by their own material payoff should not contribute to the public account and the Nash equilibrium is for no player to contribute anything. Yet, in experimental

studies, while not everyone contributes, players contribute on average 50% of their endowment in the first round (Fehr & Gächter, 2002; Fischbacher & Gächter, 2010a), but over subsequent rounds contributions decline towards zero. The decay in cooperation occurs because of the existence of two main types of players: free riders, who contribute nothing to the public account and represent a minority of players, and conditional cooperators, who contribute provided that others contribute as well and represent a much larger proportion of players (Fischbacher & Gächter, 2010b; Fischbacher et al., 2001; Isler et al., 2021; Keser & Van Winden, 2000; Kocher et al., 2008; Kopsacheilis et al., 2025; Kurzban & Houser, 2005; Thöni & Volk, 2018). Over time, as the conditional cooperators learn of the presence of the free riders, they reduce their contributions, since stopping contributing is the only way they can avoid being exploited by those players.

Thus, when cooperation depends solely upon the intrinsic motivation of players to cooperate, cooperation is fragile and undermined by the presence of free riders. However, several factors have been shown to enhance cooperation. These include the introduction of coordination devices, such as communication (Bochet et al., 2006; Brosig et al., 2003; Dawes et al., 1977; Isaac & Walker, 1988) and contribution thresholds or provision points (Isaac et al., 1989; Suleiman & Rapoport, 1988); mechanisms that enable conditional cooperators to associate, such as partner selection or endogenous group formation (Page et al., 2005); as well as extrinsic incentives such as opportunities for reputation building (Milinski et al., 2002; Semmann et al., 2004), costly monetary punishments (Fehr & Gächter, 2002; Gachter et al., 2008), and social approval (Gächter & Fehr, 1999; Rege & Telle, 2004) or disapproval (Masclet et al., 2003). The public goods game has therefore provided valuable insights into the conditions under which the provision of public goods is likely to succeed or fail, providing potential solutions for improving the provision of real-world public goods.

However, the public goods game is not an accurate experimental model of the biosecurity cooperation problem for two main reasons. First, in the public goods game,

individuals must cooperate to obtain a collective gain—an increase in the welfare of all individuals. By contrast, protection against pest invasion requires individuals to cooperate to avoid a collective loss—a reduction in the welfare of all individuals triggered by an outbreak of pests. Second, the standard game is based on a linear summation public good, where the level of provision depends on the sum of individual contributions. By contrast, protection against pest invasion has been characterised as a weakest-link public good (Burnett, 2006; Perrings, 2016; Perrings et al., 2002), where the level of provision depends not upon total contributions, but instead on the individual who contributes the least to biosecurity. In such cases, a single weak link can undermine the efforts of others, making overall protection dependent on the lowest contribution.

A prominent one-shot game-theoretic model of protection from pest invasion was introduced by Hennessy (2008). His model departs from the strict weakest-link assumption and instead considers strategic interactions between a group of beneficiaries, where individuals weigh the cost of biosecurity against the expected benefits of free-riding. Biosecurity is a risk-reduction public good in that pest freedom is risky—if a pest invades one property, it quickly spreads to all of the region's properties, reducing the welfare of everyone in the region—and contributions to the public good are concerned with reducing the risk that an incursion is established. In Hennessy's model, the public good is the joint endogenous probability of remaining pest free; conversely, the public bad is the probability of an incursion. Unlike a strict weakest-link model, the least motivated individual does not necessarily determine the overall level of provision. Instead, total contributions reduce the risk of pest invasion in a continuous, rather than binary, manner. Hennessy (2008) shows that typically the Nash equilibrium for this class of games leads to sub-optimal provision of biosecurity effort, which explains why pest prevention is an under-supplied public good associated with high levels of invasion (Burnett, 2006).

Improving biosecurity provision requires a better understanding of factors that promote or inhibit biosecurity cooperation. An economic game designed to simulate the

unique social dilemma and public goods aspects of biosecurity in the experimental laboratory could provide valuable insights. Yet, to our knowledge, no such game currently exists. In this paper, we bridge this gap in the literature by introducing a novel economic game grounded in the game-theoretic model of Hennessy (2008) that can be used to provide causal evidence on the drivers of cooperative biosecurity behaviour and their implications for policy and practice.

We chose to base our game on the Hennessy model because it is a relatively simple one-shot game-theoretic model focusing on the collective-risk management dimension of biosecurity that can be readily translated into a clear, easy-to-explain experimental paradigm. Earlier theoretical models include spatial and dynamic components of biosecurity (Hennessy, 2007; Olson & Roy, 2002, 2008). Although these features are important in many applied settings, incorporating the dynamics of an evolving pest and conveying the importance of spatial configuration in biosecurity management would introduce substantial complexity and potentially make the experimental game difficult for participants to comprehend. Such models also do not have straightforward one-shot equilibrium solutions that can be tested experimentally. For similar reasons, we opted not to use more recent theoretical models (Cobourn et al., 2019; Lampert, 2020) whose additional complexity would make them less suitable for implementation as a simple economic game.

1.2 Biosecurity collective-risk social dilemma

We refer to our game as the biosecurity collective-risk social dilemma (cf. Milinski et al., 2008) as it simulates the collective-risk management dimension that is a defining characteristic of cooperative biosecurity. The game, and by extension the Hennessy (2008) model on which it is based, applies to a variety of biosecurity situations where cooperation is required to prevent the entry of a pest into a region. However, in this paper, we frame the game in terms of the problem of a group of apple growers seeking to prevent a fruit fly outbreak.

The game involves groups of four players, each representing an apple grower in a region. It is played over multiple rounds, with each round representing a growing season. In each round, players receive a crop revenue and must pay a production cost, which is obligatory. They must also decide whether to pay an additional biosecurity cost to protect their revenue, which is voluntary. Players are informed of the risk of an outbreak of fruit fly and must decide whether to biosecure based on this risk and their beliefs about other players' strategies. Biosecurity actions reduce the risk of pest invasion in a continuous manner—if all players biosecure, the risk is eliminated. At the end of each round, the possibility of an outbreak is simulated. If no outbreak occurs, a player's payoff is their crop revenue minus production and, if applicable, biosecurity costs. If an outbreak occurs, crop revenue falls to zero, leaving players with a loss equal to their production and, where applicable, biosecurity costs.

Hennessy's model assumes that biosecurity actions depend on the expected net benefit of protection, which is jointly determined by the risk of pest entry and the number of other actors who biosecure. To test the effect of entry risk on biosecurity decisions, we compared a low-risk treatment with a high-risk treatment. We also included a no-communication treatment, in which players were unable to communicate with one another, and a communication treatment, in which players could interact via a chat room interface in certain rounds. As already noted, communication significantly increases contributions in the public goods game (Brosig et al., 2003; Dawes et al., 1977; Isaac & Walker, 1988; Sally, 1995; Zelmer, 2003). Notably, verbal communication via a chat room, like that employed in our experiment, is as effective as face-to-face communication (Bochet et al., 2006). There are many possible mechanisms by which communication may improve cooperation, and it remains unclear precisely why it is beneficial (Gächter & Herrmann, 2006). We therefore expected communication to increase biosecurity and undertook exploratory analyses of several potential mechanisms underlying this effect, which we describe below.

2 Hennessy's (2008) model of biosecurity

The augmented version of Hennessy's (2008) model used in the experiment can be described as follows. He uses a lake with boat owners as actors as his motivating example, although the model is equally applicable to horticultural pests such as fruit fly. There are multiple firms in a region. If the pest colonises one of the firms, it will immediately reduce the welfare of all firms. There is some biosecurity action, such as field inspections, spraying, or trapping, that eliminates the risk of pest incursion and establishment. All firms benefit from the absence of the pest without rivalry. Non-acting firms cannot be excluded from the benefits. Biosecurity action has a private cost, as engaging in biosecurity effort requires management time and other resources. Each firm's decision depends on what they expect the others to do.

2.1 Formal description

Formally, the model is as follows. There are N firms $i \in (1, 2, ..., N)$. Each firm derives a crop profit of $v_i - c_i^p$ and has a cost of biosecurity action of c_i^b. A firm that is affected by an incursion loses value v_i but incurs production costs c_i^p. The risk that a firm will introduce the pest are independent, identically distributed Bernoulli random variables with realisations of 0 and 1. The probability that an individual firm is not the source of an incursion is 1 if it biosecures and σ if it does not, where $\sigma \in (0, 1)$. Parameter σ can be viewed as the entry risk, where lower values of σ correspond to higher individual entry risk. It is the common baseline probability that a non-biosecuring firm does not trigger an incursion. If k firms biosecure, then the probability that the pest does not enter is σ^{N-k} (conversely, $1 - \sigma^{N-k}$ is the probability of pest entry).

The expected profit for firm i if no firms biosecure is $v_i\sigma^N - c_i^p$. If k other firms biosecure, it is $v_i\sigma^{N-k} - c_i^p$, and the expected profit of a biosecuring firm is $v_i\sigma^{N-k-1} - c_i^p - c_i^b$. The firm's objective function is:

$$\max[v_i\sigma^{N-k-1} - c_i^p - c_i^b,\ v_i\sigma^{N-k} - c_i^p] \tag{1}$$

That is, a risk neutral firm compares the expected profit from biosecuring with that from not biosecuring. Biosecuring lowers the invasion risk but incurs a cost.

The marginal private payoff to biosecuring is:

$$\Delta = v_i\sigma^{N-k-1} - c_i^b - v_i\sigma^{N-k} \tag{2}$$

This captures the gain in expected profit from acting. The condition for action is that, $\Delta>0$. By defining $\rho_i = c_i^b/v_i$, the conditions for action become:

$$\sigma^{N-k-1}(1-\sigma) > \rho_i \tag{3}$$

A firm will choose to biosecure if the expected benefit exceeds the cost. The marginal benefit of biosecuring increases with the number of other firms that biosecure. Its relationship with the entry-risk parameter is nonmonotone. As σ falls from 1, the marginal benefit of biosecuring initially increases because a non-biosecuring firm poses a greater entry risk. At sufficiently low values of σ, however, the benefit falls because an incursion is increasingly likely to occur through other non-biosecuring firms regardless of the focal firm's action. Nevertheless, as we shall see next, lower values of σ can change the strategic structure of the problem by making full biosecurity a Nash equilibrium. In this region, firms face a coordination problem—biosecuring is advantageous when others also biosecure, but may not be advantageous when they do not.

2.2 A two-firm example

Consider the case where $v_1 = v_2 = 1$, $c_1^b = c_2^b = 0.4$, and $c_1^p = c_2^p = 0.2$—which mirror the incentive structure used in the experiment—with only two firms. Table 1 presents the expected payoff calculations, while Figure 1 illustrates the expected payoffs and protection probabilities for a subset of illustrative values of the entry-threat parameter. Supplementary Figure S1 presents payoff matrices for a broader range of values of σ.

Figure 1 illustrates how variation in the entry-threat parameter changes the strategic structure of the biosecurity problem. There are two thresholds that divide the

entry-threat parameter space into three qualitatively distinct regions, each characterised by a different payoff structure. The first is the dominance threshold, defined by $\sigma = v - c^b = 0.6$. To the right of the dominance threshold ($\sigma > v - c^b$), the dominant strategy for each firm is to not biosecure and mutual non-biosecuring is the unique pure strategy Nash equilibrium. There is a second threshold, the efficiency threshold, which is sensitive to the group size N, defined by $\sigma = (1 - c^b)^{1/N}$, which equals approximately 0.77 in the two-firm example, that further divides this space.

In the region to the right of the efficiency threshold ($\sigma > (1 - c^b)^{1/N}$), the game does not involve a social dilemma: mutual non-biosecuring yields higher payoffs than mutual biosecuring and is both the individually and socially optimal outcome. The theory predicts that, in this no-dilemma region, the risk of an outbreak is sufficiently low that firms are better off not biosecuring.

Between the dominance and efficiency thresholds ($v - c^b < \sigma \leq (1 - c^b)^{1/N}$), the game is a classic prisoner's dilemma where mutual biosecuring is the socially optimal outcome, yet each firm still has an incentive not to biosecure if it expects the other firm to biosecure. The theory predicts that, in this prisoner's-dilemma region, biosecurity will be under-supplied and associated with high levels of pest outbreaks.

Finally, to the left of the dominance threshold ($\sigma \leq v - c^b$), the game is a coordination game with two symmetric pure strategy Nash equilibria: a risky equilibrium in which neither firm biosecures and a safe equilibrium in which both firms biosecure. Mutual biosecuring is the socially optimal and payoff-dominant outcome. As σ decreases within this region—and the risk of an outbreak increases—the payoff advantage of the safe equilibrium over the risky equilibrium becomes larger, potentially making mutual biosecuring increasingly *focal* or psychologically prominent (Schelling, 1980). The theory predicts that, in this coordination-game region, the prospect of cooperation should be greater than in the prisoner's-dilemma region because mutual biosecuring is now a Nash equilibrium. Nevertheless, mutual non-biosecuring remains a Nash equilibrium, so there is

still a risk of coordination failure if each firm is uncertain about the intentions of the other.

2.3 Parameter values and predictions

For the experiment, which involved four-player groups, we chose values of $\sigma = 0.9$ (low risk) and $\sigma = 0.8$ (high risk) for our risk manipulation. In the four-player case, the efficiency threshold is $\sigma = (1 - c^b)^{1/N} \approx 0.88$. Thus, $\sigma = 0.9$ falls within the no-dilemma region, where mutual non-biosecuring is both individually and socially optimal, whereas $\sigma = 0.8$ falls within the prisoner's-dilemma region, where non-biosecuring is individually rational but mutual biosecuring is socially optimal. The prisoner's-dilemma region is of particular interest because it is associated with the greatest risk of biosecurity being under-provided, whereas the no-dilemma region provides a baseline for examining behaviour in the absence of a social dilemma.

The theory predicts mutual non-biosecuring in the two risk treatments, as not biosecuring is the dominant strategy and the unique Nash equilibrium in both cases. However, we know that the degree of cooperation in economic games is greater than predicted by standard economic theory. For example, experimental studies of the prisoner's dilemma find that a considerable fraction of players prefer to cooperate rather than defect (Kiyonari et al., 2000; Sally, 1995) and, as we have already seen, in the public goods game, which is an N-person prisoner's dilemma, a substantial number of players contribute to the public good (Chaudhuri, 2011; Ledyard, 1995). Even in the low-risk treatment, players may misperceive the payoff structure and think that some degree of cooperation is warranted. Alternatively, some players may have risk preferences that make biosecuring individually optimal (see Supplementary Game-Theoretic Analysis). Nevertheless, because mutual biosecuring is socially optimal in the high-risk treatment but not in the low-risk treatment, we predicted that players would be more likely to biosecure in the high-risk than the low-risk treatment.

Of critical interest is how the presence versus absence of communication influences biosecurity decisions. Communication affords the opportunity for players to provide

assurances to one another that they will cooperate if others are prepared to do so too. If each player is concerned not only about their own payoff, but also that of their co-players, then communication may turn the prisoner's dilemma payoff structure into an assurance game where mutual biosecuring and mutual non-biosecuring are both Nash equilibria. In other words, communication may turn the prisoner's dilemma into a coordination game like that which exists when $\sigma \leq v - c^b$. Which of these Nash equilibria players will coordinate on depends on each players beliefs about what the other players will do.

We predicted that players would be more likely to biosecure when communication was available than when it was absent. We also expected an interaction between the risk and communication manipulations—specifically, that communication would increase the likelihood of biosecuring more in the high-risk than the low-risk treatment.

3 Beyond the Hennessy (2008) model

Beyond testing aspects of the Hennessy model, we sought to address several additional questions about the factors influencing biosecurity decisions in the biosecurity collective-risk social dilemma.

First, because players in our experiment received feedback at the end of each round about the biosecurity decisions of their group members (social feedback) and whether an outbreak of pests had occurred (outcome feedback), we asked whether these sources of feedback influenced biosecurity decisions on the subsequent round. We were also interested in whether the effects of these two sources of feedback were moderated by the risk level and the presence or absence of communication.

There is a wealth of evidence from the public goods game that many people are conditional cooperators who are willing to contribute to the public good provided that others are willing to do so as well (Fischbacher & Gächter, 2010b; Fischbacher et al., 2001; Isler et al., 2021; Keser & Van Winden, 2000; Kocher et al., 2008; Kopsacheilis et al., 2025; Kurzban & Houser, 2005; Thöni & Volk, 2018). If conditional cooperation is an important strategy influencing biosecurity decisions, then social feedback should have a positive

effect—individual players should be more likely to biosecure as a function of the proportion of other group members who chose to biosecure in the previous round.

Whereas social feedback was expected to influence biosecurity decisions through conditional cooperation, outcome feedback may influence biosecurity decisions by providing a noisy but informative signal regarding the collective adequacy of biosecurity. The occurrence of an outbreak might be interpreted by players as evidence that the current level of biosecurity is collectively insufficient. If such outcome-dependent adaptation of player strategies influences biosecurity decisions, then outcome feedback should have a positive effect—individual players should be more likely to biosecure following the occurrence of an outbreak on the previous round.

Second, we classified participants according to their cooperation type to determine if the presence of communication changed the proportion of players exhibiting different biosecurity decision profiles compared to when communication was unavailable. In the public goods game, player types are typically classified using the "strategy method", whereby participants indicate how much they would contribute to the public good contingent upon different average contribution levels of other players (Fischbacher et al., 2001). This method has identified several different types of players, with the most prominent distinction being between free riders (who always contribute nothing) and conditional cooperators (who closely match others' contributions) (see e.g., Fischbacher et al., 2001). As noted earlier, conditional cooperators typically represent a larger proportion of participants than free riders (Fischbacher & Gächter, 2010b; Fischbacher et al., 2001; Kocher et al., 2008). For example, based on an analysis of 17 experiments using the strategy method, Thöni and Volk identified approximately 61% of participants as conditional cooperators and 19% as free riders. In our experiment, we adopted a different approach and classified players based on their in-game behaviour using their responses on rounds in which at least two of the other three players had biosecured in the previous round. Our classification scheme distinguished between conditional cooperators, potential

free riders, and mixed responders. It should be noted that there are limitations to this approach. Unlike the strategy method, our approach classifies a player's type based on their responses to the range of decisions of other group members that happens to occur during the game, rather than eliciting their responses to a systematic range of possible decisions of other group members. Consequently, the classification of a player may depend on the conditions they experience and is unlikely to reflect their underlying social preferences.

Third, we asked whether fundamental economic preferences—namely risk aversion and loss aversion—were related to biosecurity decisions and, if so, whether these relationships varied according to the risk level and the presence or absence of communication. To address this question, our study included a virtual lottery game (Tanaka et al., 2010) administered prior to the main experiment in which participants' preferences over risky gambles could be used to estimate the values of parameters representing risk aversion and loss aversion using cumulative prospect theory (Tversky & Kahneman, 1992). Complementing these objective, parameter-based measures, the post-experiment survey additionally included subjective self-report measures of these two constructs.

Fourth, we asked whether communication increased trust and fostered changes in player strategies. One reason why communication may increase cooperation is that it allows the exchange of commitments to cooperate (Bochet et al., 2006; Brosig et al., 2003). Yet, for communication to change expectations of others' behaviour, mutual trust in the commitments exchanged must increase (Ostrom, 2003). Trust may therefore be a key psychological process underpinning the effectiveness of communication. To test whether communication increased trust, the post-experiment survey included self-report measures of general trust—the dispositional tendency to trust others—situational trust towards other group members in the experimental game, and strategy use. Of particular interest is whether communication increased situational trust and the adoption of conditionally cooperative strategies.

Finally, we asked how communication was used to coordinate group behaviour. To address this question, we undertook computational text mining and sentiment analyses of the content of group communication. Previous studies of communication in public goods games, and other types of social dilemmas, have revealed that it is often used to discuss the joint optimal strategy, agree on what everyone should contribute, and express social disapproval towards players that renege on their commitments (Bochet et al., 2006; Brosig et al., 2003; Isaac & Walker, 1988; Ostrom & Walker, 1997; Ostrom et al., 1994). We expected our analysis to uncover a similar use of communication to coordinate attempts to secure cooperation.

4 Experiment

Ethical approval to undertake the research was granted by the Human Ethics office at the [redacted].

4.1 Participants

Ninety-six participants (50% females; mean age = 20.72, SD = 5.26, range = 18 – 23) were recruited from the campus community at the [redacted]. Participants were recruited using the Online Recruitment System for Experimental Economics (ORSEE; Greiner, 2015).

4.2 Design

The experiment employed a 2 (risk: low vs. high) × 2 (communication: no communication vs. communication) × 15 (round: 1–15) mixed design: risk and round were within-groups factors, whereas communication was a between-groups factor. Participants were tested in groups of four players. Each group played two 15-round blocks, one under each risk treatment, with block order counterbalanced across groups. Groups were allocated randomly and evenly to the no-communication and communication treatments, resulting in 12 groups for each level of the communication factor.

4.3 Apparatus and procedure

Experimental sessions were conducted in the [redacted], in the presence of the experimenter. At the start of a session, players were randomly seated at interconnected computer terminals running the Zurich Toolbox for Readymade Economic Experiments (z-Tree; Fischbacher, 2007), which was used to register and communicate their decisions during the experiment. The computer terminals were separated by privacy blinds to prevent player collusion. Participants read an information sheet before providing written informed consent. Sessions comprised three parts: a virtual lottery game, the biosecurity collective-risk social dilemma, and a post-game questionnaire.

4.3.1 Ex ante virtual lottery game

Participants first completed a virtual lottery game (Supplementary Virtual Lottery Game) taken from Tanaka et al. (2010). They were required to choose between playing for the monetary amounts under two different prizes. There were 35 games in total, separated into three play panels of 15, 15, and 5 games. For each game, players were informed of the odds of each prize and prompted to select their preferred prize. The game was included to permit estimation of participants' degree of risk aversion and loss aversion, using cumulative prospect theory (Tversky & Kahneman, 1992), so these estimates could be related with their decisions in the biosecurity collective-risk social dilemma.

4.3.2 Biosecurity collective-risk social dilemma

Next, participants read the experimental instructions for the biosecurity collective-risk social dilemma (Supplementary Experimental Instructions). The instructions explained to participants that they would be participating in a collective decision-making game related to risky choice with three other players. They were informed that, in this game, they owned an apple orchard and were surrounded by three neighbours who also grew apples. They were told that each round of the game represented an apple-growing season and that seasonal revenues were earned on those rounds where no outbreak of pests

occurred. If an outbreak occurred, the pests spread quickly and infested all players' orchards, resulting in the loss of that season's revenue. Players were informed that they would have to decide whether to protect (at a cost) or not protect (at no cost) against a pest outbreak at the start of each season. They knew that if all players protected, the pest would be prevented with certainty. Once players had finished reading the instructions, they answered a series of control questions to ensure that they understood the rules of play (Supplementary Control Questions). When the experimenter was satisfied that the questions had been answered correctly, the game commenced.

At the start of the game, each player was assigned a pseudonym. During the game, each player's decisions were communicated to the other players under their designated pseudonym. The procedure for the no-communication treatment was as follows. Each round consisted of a biosecurity-decision stage and a feedback stage. At the start of the biosecurity-decision stage, each player was endowed with the seasonal revenue generated by their orchard ($v = \$25$). This amount represented the gross value of the crop for that season. The decision they had to make—simultaneously and independently—was whether to protect or not protect their orchard against an outbreak of pests, where protection incurred a cost ($c^b = \$10$). Players also knew that, regardless of whether they chose to protect or not, a seasonal production cost ($c^p = \$5$) would be deducted when calculating their realised payoff for the round. Once all decisions were made, the possibility of an outbreak was simulated. Following Hennessy (2008), the probability of pest entry was calculated as:

$$p = 1 - \sigma^{N-k} \tag{4}$$

where N is the total number of players, k is the number of players deciding to protect, and σ is the entry risk parameter, which was set to 0.9 and 0.8 in the low-risk and high-risk treatments, respectively. Table 2 shows how these values of σ translate into outbreak probabilities in the two risk treatments, as a function of the number of players

choosing to protect. Players were fully informed of these probabilities before the game began. As shown in Table 2, each additional player choosing to protect reduced the probability of a pest outbreak, with an outbreak prevented with certainty when all players protected. A random number $r \in (0, 1)$ was subsequently drawn from a uniform distribution to determine whether an outbreak occurred. If $r \leq p$, then an outbreak occurred and all players lost that season's revenue v; otherwise, an outbreak was prevented and all players retained that season's revenue, minus any costs incurred. The four possible net payoff outcomes for each round are summarised in Table 3.

In the feedback stage, players received information about the biosecurity decisions made by each player and whether an outbreak of pests had occurred. They were also told their net payoff for the current round and the balance of their operating account, which represented the cumulative sum of net payoffs obtained across all rounds completed so far in the current block. Upon entering the second block, players' operating accounts were reset to zero. Players were informed that their previously accrued earnings were not lost and would be paid at the end of the experiment together with the net payoff accumulated in the forthcoming block.

In the communication treatment, the procedure was the same as described above, except that before the biosecurity-decision stage on the first, sixth, and eleventh rounds of both the low-risk and high-risk treatments, players had the opportunity to communicate with each other via a chat terminal. Conversations lasted for two minutes, with a countdown timer displaying the amount of time remaining, and during that time players were free to discuss task-related and task-unrelated topics. Chatroom conversations were monitored by the experimenter, and players were instructed not to use abusive language or offer side payments to induce cooperation.

4.3.3 Ex post questionnaire

Experimental sessions concluded with a brief questionnaire. Two self-report measures of risk and loss aversion were included to compare with the estimates from the

virtual lottery game. To assess risk aversion, participants were asked to "Imagine you have $40. With a probability of 50%, you will lose all $40. You can avoid the risk by giving away $20 of the $40. Would you pay the $20 to avoid the risk?". Responses (0 = no, 1 = maybe, 2 = yes) were used to classify participants as risk-seeking, risk-neutral, or risk-averse. To assess loss aversion, participants rated how well the statement "I generally hate to lose something more than I like to gain something" described them (1 = does not describe me at all, 7 = describes me perfectly).

Participants also completed general and situational trust measures. General trust was measured using the Generalized Trust Scale (T. Yamagishi & Yamagishi, 1994). The scale contains six-items (e.g., "Most people are trustworthy") and participants indicated the extent to which they agreed or disagreed with each statement (1 = strongly disagree, 5 = strongly agree). Situational trust was measured by asking participants how much they trusted each group member (1 = no trust, 5 = a lot of trust).

To identify player strategies, participants were asked to indicate which of six different factors was most important to their decision of whether to protect or not: (1) "I thought that other players would not protect, so I didn't see why I should either" (non contributor); (2) "I thought others would protect and I could benefit without having to protect myself" (free rider); (3) "I tried to avoid incurring monetary losses" (loss averse); (4) "I thought the other players would choose to protect, so I protected as well" (conditional cooperator); (5) "I thought that choosing to protect was the right thing to do, regardless of what other players decided" (moral cooperator); (6) "I didn't know what others were going to do and thought it was worth the risk of taking protective action" (worthwhile cooperator).

Experimental sessions lasted approximately 60 minutes. At the end of each session, participants were paid a $5 show-up fee and the outstanding balances of their operating accounts in cash. A 10% conversion was applied to the experimental profits to calculate the actual cash payments. For example, a player who earned $195 would receive $19.50. This

conversion rate was explained in advance.

5 Results

The results are organised into five sections. First, we examine the relation of the experimental manipulations of risk, communication, and round, as well as prior-round feedback, to biosecurity decisions. Second, we assess the degree to which behaviour conformed to or departed from two key theoretical benchmarks: the Nash equilibrium and cooperative outcome. Third, we examine the distribution of different player types and whether this changed with the availability of communication. Fourth, we analyse the relation of the economic-preference measures (risk aversion and loss aversion) and biosecurity decisions. We then compare the no-communication and communication treatments on the self-report measures of trust and strategy use to determine whether, and how, these responses varied as a function of communication. Finally, we undertake a computational text analysis of the contents of communication to determine how it was used to coordinate behaviour. All statistical analyses were undertaken using R Statistical Software (v4.4.0; R Core Team, 2024).

5.1 Biosecurity decisions

Biosecurity decisions were modelled using generalized linear mixed-effects models, with a binomial distribution and a logit link function, implemented employing the *lme4* package in R using maximum likelihood parameter estimation. The dependent variable was an individual player's biosecurity decision on a given round (1 = protect, 0 = not protect). This approach allows us to model the binary nature of responses, while accounting for the repeated-measures structure of the data, with decisions nested within players, and players nested within groups, thus respecting the true hierarchical organisation of the data.

To test the Hennessy model, we begin with a treatment-only model incorporating the factors manipulated in the experiment but excluding the effects of prior-round feedback, which we consider later. The model included the full factorial structure of the experimental design, comprising the main effects of risk, communication, and round, along

with all corresponding two-way and three-way interactions between these predictors. To determine a suitable random effects structure, we compared various models with different combinations of random effects at the group and player level and used the Akaike Information Criterion (AIC; Akaike, 1974) and Bayesian Information Criterion (BIC; Schwartz, 1978) to select the preferred model (Supplementary Statistical Analyses: Selection of Random-Effects Structure). This approach of using model selection criteria to select a random effects structure that is supported by the data confers the advantage that it preserves statistical power without inflating Type I error rates (Matuschek et al., 2017). The preferred model included random intercepts for both group and player, a by-group random slope for round, and by-player random slopes for risk and round, as well as the correlations among these parameters. Formally, the preferred model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{5}$$

Table 4 presents a summary of the treatment-only model estimates for the fixed effects (including random effects estimates and model fit). Figure 2 shows the observed proportion of players choosing to protect (top panels) and model estimates of the predicted probability of protecting (bottom panels), across the fifteen rounds, with and without communication, in the low-risk (left panels) and high-risk (right panels) treatments.

Starting with the main effects, there was a significant main effect of risk, $\beta = 1.88$, $SE = 0.28$, $z = 6.76$, $p < .001$, reflecting a higher probability of protecting under high risk than low risk, a significant main effect of communication, $\beta = 2.22$, $SE = 0.69$, $z = 3.22$, $p = .001$, with a higher probability of protecting in the presence than in the absence of communication, and a significant main effect of round, $\beta = -0.62$, $SE = 0.28$, $z = -2.24$,

$p = .025$, indicating that the probability of protecting decreased over rounds.

Turning to the two-way interactions, there was a significant risk × communication interaction, $\beta = 0.78$, $SE = 0.38$, $z = 2.05$, $p = .041$, indicating that the effect of communication was stronger under high risk than low risk. The risk × round interaction was not significant, $\beta = -0.22$, $SE = 0.17$, $z = -1.26$, $p = .207$, nor was the communication × round interaction, $\beta = 0.50$, $SE = 0.38$, $z = 1.31$, $p = .191$. Thus, although the probability of protecting decreased over rounds overall, there was no clear evidence that this trend varied reliably as a function of risk or communication.

Finally, the three-way interaction between risk, communication, and round was not significant, $\beta = 0.06$, $SE = 0.24$, $z = 0.24$, $p = .808$.

We next explore whether biosecurity decisions were influenced by feedback from the previous round. To do this, we created two lagged predictors. The first, *lagged protection*, was defined as the proportion of other group members who protected in the preceding round and represents social feedback about others' behaviour. The second, *lagged outcome*, was defined as whether an outbreak occurred in the preceding round and represents feedback about realised consequences. It is important to note that because outbreaks are less likely when more players protect, these two lagged predictors are naturally correlated with one another. We fit three models and used the AIC and BIC to select the preferred model. Note that because lagged predictors are not defined for the first round, the feedback models were estimated using data from Rounds 2–15. Each model included the same predictors as the treatment-only model, including the same random effects structure, but they differed in terms of how the lagged predictors were incorporated.

The first model, dubbed the additive-feedback model, included lagged protection and lagged outcome as additive predictors, assuming that responsiveness to prior feedback does not vary across treatments. Formally, the additive-feedback model was specified as:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_protection} + \text{lagged_outcome} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{6}$$

The second model, dubbed the social-feedback model, allowed the effect of lagged protection to vary as a function of risk and communication, permitting an assessment of whether responsiveness to others' prior biosecurity decisions varies by treatment. To determine the appropriate model specification, we fit two models. Both models included the two-way interactions of lagged protection with risk and communication. In one model, we also incorporated the three-way interaction between these predictors. Although the AIC favoured the three-way interaction model, the BIC favoured the more parsimonious two-way interaction model (Supplementary Statistical Analyses: Social-Feedback Model Selection) and it was this model that we chose. Formally, the preferred model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_outcome} + \text{lagged_protection} \\ & + \text{risk} : \text{lagged_protection} \\ & + \text{communication} : \text{lagged_protection} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{7}$$

The third model, dubbed the outcome-feedback model, instead allowed the effect of lagged outcome to vary as a function of risk and communication, permitting an assessment of whether responsiveness to prior outbreak occurrence varies by treatment. As per the social-feedback model, to determine the appropriate model specification, we fit two models. Both models included the two-way interactions of lagged outcome with risk and communication. In one model, we also incorporated the three-way interaction between these predictors. Both the AIC and BIC unambiguously favoured the two-way interaction model (Supplementary Statistical Analyses: Outcome-Feedback Model Selection) and it was this model that we chose. Formally, the preferred model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_protection} + \text{lagged_outcome} \\ & + \text{risk} : \text{lagged_outcome} \\ & + \text{communication} : \text{lagged_outcome} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{8}$$

Table 5 presents the goodness-of-fit statistics for the feedback models including a version of the baseline treatment-only model fitted to Rounds 2–15 to permit comparison. It can be seen that all three feedback models provided a better fit than the treatment-only model, as indicated by substantially lower AIC and BIC values. A likelihood ratio test (LRT) confirmed that the additive-feedback model provided a significant improvement in fit over the treatment-only model, $\chi^2_{df=2} = 82.57$, $p < .001$. The preferred model, indicated by the lowest AIC and BIC, was the social-feedback model in which lagged protection varied by treatment. The superiority of the social-feedback model was further supported by LRTs showing that it provided a significant improvement in fit over the additive-feedback

model, $\chi^2_{df=2} = 20.01$, $p < .001$, whereas the outcome-feedback model did not provide a significant improvement in fit over the same baseline model, $\chi^2_{df=2} = 3.80$, $p = .150$. Accordingly, we interpret our results in terms of the social-feedback model.

A summary of the social-feedback model estimates for the fixed effects is presented in Table 6 (including random effects estimates and model fit). Starting with the lagged predictors, there was no significant main effect of lagged protection, $\beta = 0.71$, $SE = 0.54$, $z = 1.32$, $p = .185$. There was also no significant main effect of lagged outcome, $\beta = 0.17$, $SE = 0.14$, $z = 1.26$, $p = .207$, indicating that the probability of protecting was unaffected by whether or not an outbreak had occurred in the previous round.

Critically, responsiveness to social feedback varied as a function of communication. There was a significant communication $\times$ lagged protection interaction, $\beta = 2.47$, $SE = 0.56$, $z = 4.40$, $p < .001$. The interaction, plotted in Figure 3, arose because the probability of protecting increased as a function of the proportion of other players that protected in the previous round, but the association was stronger in the presence than in the absence of communication. The risk $\times$ lagged protection interaction was marginal, $\beta = 0.86$, $SE = 0.50$, $z = 1.72$, $p = .085$, reflecting a weaker trend for social feedback to exert a stronger positive effect on the probability of protecting under high risk than low risk. Thus, communication increased responsiveness to social feedback, whereas the effect of risk on responsiveness to others' prior biosecurity decisions was weaker.

After controlling for the effects of prior-round feedback, the main effect of risk remained significant, $\beta = 1.19$, $SE = 0.34$, $z = 3.47$, $p < .001$, with a higher probability of protecting under high risk. The main effect of communication did not reach conventional levels of significance, $\beta = 0.88$, $SE = 0.60$, $z = 1.48$, $p = .138$. The risk $\times$ communication interaction was also not significant, $\beta = -0.42$, $SE = 0.42$, $z = -0.98$, $p = .327$, indicating that once responsiveness to social feedback was taken into account, there was no reliable evidence of an additional combined effect of risk and communication on the probability of protecting.

Finally, there was a significant main effect of round, $\beta = -0.64$, $SE = 0.20$, $z = -3.27$, $p = .001$, with the probability of protecting decreasing over rounds. The communication $\times$ round interaction was marginal, $\beta = 0.47$, $SE = 0.25$, $z = 1.85$, $p = .064$, while the risk $\times$ round interaction, $\beta = -0.02$, $SE = 0.19$, $z = -0.12$, $p = .901$, and the three-way interaction with round, $\beta = -0.04$, $SE = 0.26$, $z = -0.15$, $p = .878$, were not significant.

5.2 Benchmark comparisons

We next examine whether rates of protection conformed to or departed from key theoretical benchmarks. We averaged biosecurity decisions at the group level within each treatment and compared these means against 0, corresponding to the Nash equilibrium, and 1, corresponding to the cooperative outcome, using one-sample t-tests. The data are portrayed in Figure 4 which shows the proportion of players choosing to protect, with and without communication, in the low-risk and high-risk treatments without respect to round.

In the low-risk treatment, the Nash equilibrium is both the individual and socially optimal outcome. In the absence of communication, rates of protection were closest to, yet still significantly above, the Nash equilibrium, $t(11) = 3.01$, $p = .012$, whereas with communication, protection rates were considerably higher than this benchmark, $t(11) = 5.90$, $p < .001$. In both cases, protection rates were markedly below the cooperative outcome, although the difference was larger without communication, $t(11) = 12.40$, $p < .001$, than with communication, $t(11) = 7.27$, $p < .001$.

By contrast, in the high-risk treatment, the cooperative outcome is the socially optimal outcome. In the absence of communication, rates of protection were considerably below the cooperative outcome, $t(11) = 7.47$, $p < .001$, whereas with communication, protection rates were closer to, yet still significantly below, this benchmark, $t(11) = 4.81$, $p < .001$. In both cases, protection rates were significantly above the Nash equilibrium, although the difference was larger with communication, $t(11) = 20.60$, $p < .001$, than without communication, $t(11) = 4.83$, $p < .001$.

5.3 Player types

Turning to the classification of player types, for this analysis we restrict our focus to the data for the high-risk treatment. This is because, in the low-risk treatment, the individually and socially optimal outcome is for players not to biosecure. Some players may be aware of this fact, whereas others may be naive to it. This becomes problematic when classifying potential free riders: a player who does not protect may do so because they are free-riding on the protective actions of others or because they recognise that protecting is unnecessary under these conditions. Since we cannot distinguish between these two possibilities, undertaking the player-type analysis on the low-risk treatment data could be potentially misleading.

We classified players based on their responses in rounds in which at least two of the other three players had protected in the previous round. Players who protected on at least 75% of such opportunities were classified as conditional cooperators, those who protected on 25-75% as mixed responders, and those who protected on 25% or fewer as potential free riders. Players with no eligible opportunities were excluded from this classification. The classification criteria are somewhat arbitrary but distinguish players according to whether their behaviour is predominantly cooperative, non-cooperative, or mixed.

Using these criteria, in the no-communication treatment, 15 of 45 players (33%) were classified as conditional cooperators, 11 (24%) as mixed responders, and 19 (42%) as potential free riders. In the communication treatment, by comparison, 40 of 48 players (83%) were classified as conditional cooperators, 8 (17%) as mixed responders, and none as potential free riders. Accordingly, in the absence of communication, potential free riders constituted the largest group, whereas in the presence of communication the large majority of players were classified as conditional cooperators and no players were classified as potential free riders.

5.4 Risk aversion, loss aversion, trust, and player strategies

We now examine whether individual players' biosecurity decisions were affected by their degree of risk aversion and loss aversion, before exploring whether communication influenced their propensity to trust others and the strategies they adopted.

We begin by examining the association between each individual player's biosecurity decisions and their level of risk and loss aversion, as indexed by their preferences in the virtual lottery game, and their responses on the self-report items. For the analysis based on the virtual lottery game, players' prize preferences in the three play panels were used to estimate the value of two parameters, using an augmented version of cumulative prospect theory (Tversky & Kahneman, 1992) incorporating the one parameter form of Prelec's (1998) weighting function (for a description of the parameter estimation procedure, see Tanaka et al., 2010). The parameters from the original paper were σ and λ, representing the curvature of the value function and the degree of loss aversion, respectively ($1 - \sigma$ measures relative risk aversion for gains, whereas λ measures loss aversion).

The relation between these economic preference indicators and biosecurity decisions was examined using a generalised linear mixed-effects model. The model included the main effects of risk and communication, and the interaction between them (excluding the effects of round for simplicity), in addition to the main effects of risk aversion and loss aversion, and their two-way interactions with risk and communication. Random intercepts were also included for both group and player, as well as a by-player random slope for risk. Two versions of the model were fitted, one using the self-report measures and another using the parameter-based measures. Formally, the models were specified as:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{risk} : \text{communication} \\ & + \text{risk_aversion} + \text{loss_aversion} \\ & + \text{risk_aversion} : \text{risk} + \text{risk_aversion} : \text{communication} \\ & + \text{loss_aversion} : \text{risk} + \text{loss_aversion} : \text{communication} \\ & + (1 \mid \text{group}) + (1 + \text{risk} \mid \text{player}) \end{aligned} \tag{9}$$

Summaries of the fixed-effect estimates for the self-report and parameter-based models are presented in Supplementary Tables S7 and S8, respectively. For both models, there were significant main effects of risk and communication, mirroring the general pattern observed in the treatment-only model reported earlier. The risk × communication interaction was significant in the self-report model and marginal in the parameter-based model.

There was no evidence that loss aversion influenced biosecurity decisions. The main effect of loss aversion (self-report model: $\beta = -0.05$, $SE = 0.20$, $z = -0.24$, $p = .812$; parameter-based model: $\beta = 0.15$, $SE = 0.19$, $z = 0.75$, $p = .454$), as well as its interactions with risk (self-report model: $\beta = 0.13$, $SE = 0.16$, $z = 0.81$, $p = .417$; parameter-based model: $\beta = -0.12$, $SE = 0.17$, $z = -0.69$, $p = .492$) and communication (self-report model: $\beta = -0.07$, $SE = 0.25$, $z = -0.27$, $p = .790$; parameter-based model: $\beta = -0.06$, $SE = 0.25$, $z = -0.26$, $p = .799$), were all not significant.

By contrast, both models showed a significant main effect of risk aversion (self-report model: $\beta = 0.61$, $SE = 0.22$, $z = 2.75$, $p = .006$; parameter-based model: $\beta = 0.64$, $SE = 0.20$, $z = 3.19$, $p = .001$), as well as a significant risk aversion × communication interaction (self-report model: $\beta = -0.60$, $SE = 0.26$, $z = -2.34$, $p = .019$; parameter-based model: $\beta = -0.55$, $SE = 0.25$, $z = -2.19$, $p = .029$). As shown in Figure 5, the interaction arose because, in the absence of communication, the probability of protecting increased as a function of risk aversion, whereas in the presence of

communication, it was largely insensitive to risk preferences.

There was no evidence that the association between risk aversion and biosecurity decisions differed by risk level. The risk × risk aversion interaction was not significant in either the self-report model, $\beta = 0.03$, $SE = 0.16$, $z = 0.20$, $p = .840$, or the parameter-based model, $\beta = -0.22$, $SE = 0.17$, $z = -1.31$, $p = .192$.

Turning to trust, Figure 6 shows average player responses on the general (top-left panel) and situational (top-right panel) trust measures for the no communication and communication treatments. Communication significantly increased both types of trust, albeit more so for situational trust, $t(94) = –4.39$, $p < .001$, than general trust, $t(94) = –3.19$, $p < .01$.

Finally, the strategies adopted by players are shown in the bottom panel of Figure 6. The dominant strategy adopted by players in both the no-communication and communication treatments—occurring in roughly equal proportions—was loss aversion. The proportion of players adopting the free rider, moral cooperator, and worthwhile cooperator strategies was comparable in the two treatments. However, the communication treatment was associated with a decrease in the non contributor strategy and an increase in the conditional cooperator strategy. This suggests that in the absence of communication, players had low expectations of cooperation from their group members, whereas in the presence of communication those expectations were increased.

To determine whether there were significant differences between the two treatments in their strategy choices, a series of Fishers Exact Tests were performed. There was a significant difference between no-communication and communication treatments for the non-contributor strategy, $p = .006$, and the conditional-cooperator strategy, $p < .001$. All other comparisons were non-significant, $p = 1.000$ for all tests.

5.5 Contents of communication

To understand how communication improved cooperation, we undertook a text mining and sentiment analysis of the contents of group conversations in R (packages used:

snowball, *syuzhet*, *tidytext*, *tidyverse*, *tm*, and *wordcloud*).

We imported the data and cleaned the text by removing special characters and whitespace before converting it to lower case. We then removed stop words (e.g., "the", "and") and stemmed words to their root forms.

Next, we constructed a word-frequency matrix. The top-left panel of Figure 7 shows a word cloud of the 100 most frequent words, with size indicating frequency. The top five words were "protect" (226), "round" (90), "risk" (73), "good" (58), and "yes" (45).

We used the AFFIN lexicon (Nielsen, 2011) to assign each word a sentiment score ranging from –5 (negative) to +5 (positive). The top-right panel of Figure 7 shows a sentiment-coded word cloud of the 100 most frequent words, with positive words in grey and negative words in black. It can be seen that the words "risk" and "outbreak" were the strongest contributors to negative sentiment, while "protect" contributed most to positive sentiment. Despite the higher concentration of negative words, the median sentiment score for the corpus was positive (1).

To understand the context of the top three words, we conducted a word-association analysis using a .15 correlation threshold to capture the most salient associations. "Protect" was not associated with any word, indicating that the context in which it was used was highly variable. Manual inspection revealed that it was largely used as a declaration to protect, to protect in a given situation (e.g., for a given risk level or round), or in the context of asking whether the group should protect, or whether a particular group member would protect.

"Risk" was strongly associated with "low" (.63) and "high" (.53), and more weakly with "part" (.21) and "tage" (.21)—terms referring to the low-risk and high-risk blocks. These associations arose because group conversations, especially at the start of each block, focused on the risk level. Groups recognised that different strategies were required in the low-risk and high-risk treatments, and the first round of communication in each block involved groups negotiating what actions should be taken, given the risk level. "Risk" also

had associations with "take" (.3) and "reckon" (.24). These associations arose because statements involving the word "risk" were often made in the context of deciding whether the group thought ("*reckon*ed") they should "take" action to protect.

"Round" had associations with "two" (.22), "everi" (the root of "every"; .22), and "person" (.18), reflecting a common strategy amongst groups, which entailed rotating protection over rounds in pairs, such that two players would protect, and two would not protect, on one round, before switching on the next. Variants of this strategy involved rotating protection in one's (low-risk treatment) and in three's (high-risk treatment). Another strategy in the high-risk treatment, again reflected in the association between "round" and "every", was to protect on every round. "Risk" also had associations with "first" (.16) and "next" (.17), which reflects the efforts of groups to coordinate who would protect "first" or "next" in the subsequent set of rounds, given use of an alternation strategy. Finally, "round" was associated with "last" (.29). This association arose in the context of announcements that over the forthcoming set of rounds, groups should repeat what they did over the "last" set of rounds.

Lower frequency negative words (e.g., "nasty", "traitor", "trick"), appeared when group members criticised players who had reneged on their commitments to protect. These instances were rare—most participants honoured their commitments. Lower frequency positive words (e.g., "cool", "glad", "happy") were associated with celebrations of successful group outcomes.

Finally, we used the NRC Emotion Lexicon (Mohammad & Turney, 2010; Nielsen, 2011) to classify words into eight emotion categories (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust). Figure 8 shows the proportion of words falling into each emotion category. Anticipation had the highest proportion of words (22.5%), followed closely by trust (18.5%). This meshes well with our earlier results demonstrating that communication facilitates the building of trust between group members. Disgust had the smallest proportion of words (5.4%). Positive emotions (trust and joy) accounted for 32% of

words, whereas negative emotions (disgust, fear, and sadness) accounted for 21% of words.

In sum, conversations revealed that groups used communication to set goals and coordinate strategies. While some naming and shaming occurred when players broke commitments, emotion analysis suggests trust was relatively high overall.

6 Discussion

In this paper, we introduced a novel economic game—the biosecurity collective-risk social dilemma—for simulating the problem of collective pest management. The game is a repeated version of the one-shot game-theoretic model of Hennessy (2008), which assumes risk neutrality, and serves as an experimental vehicle for testing its assumptions and examining the broader factors influencing self-organisation and cooperation in biosecurity contexts. According to the model, biosecurity decisions depend on the expected net benefit of protection, which varies with both the risk of pest entry and the extent to which others also take protective action. Using the game, we tested this prediction by comparing behaviour in a low-risk and a high-risk treatment, and examined whether the presence versus absence of communication influenced biosecurity decisions under these two risk scenarios.

We also addressed several questions that extend beyond Hennessy's one-shot model. Specifically, we examined whether biosecurity decisions were sensitive to prior-round feedback about other group members' protection decisions (social feedback) and/or whether an outbreak occurred (outcome feedback), and if responsiveness to either or both sources of feedback interacted with the risk and communication manipulations. Additionally, we investigated the relationships of risk and loss aversion to biosecurity decisions, compared player types, trust, and self-reported strategy use with and without communication, and analysed group conversations to understand how communication was used to coordinate collective action.

6.1 Overview of key findings

Biosecurity decisions were sensitive to both the risk level and the availability of communication. The probability of biosecuring was greater when the entry risk was high than when it was low, and in the presence versus absence of communication. The latter finding is consistent with experiments using the public goods game showing that communication enhances cooperation (Brosig et al., 2003; Dawes et al., 1977; Isaac & Walker, 1988; Sally, 1995; Zelmer, 2003). Additionally, when prior-round feedback was excluded, the effect of communication was stronger under high risk than low risk. The communication analyses suggest that this result occurred because groups recognised that greater protective action was required under high risk and used communication to coordinate on a higher cooperation level.

The benchmark comparisons revealed systematic departures from equilibrium predictions. Group biosecurity rates were reliably above the Nash equilibrium but below the cooperative outcome in all treatments. In the high-risk treatment, the cooperative outcome is the socially optimal outcome, whereas in the low-risk treatment the Nash equilibrium is both individually and socially optimal. Biosecurity was therefore underprovided under high risk, but overprovided under low risk. Communication helped overcome the underprovision problem under high risk, but worsened the overprovision problem under low risk. Thus, communication did not consistently move groups towards the socially optimal outcome.

The feedback analyses provide insights into the temporal and social dynamics underlying the above effects. Biosecurity decisions were unaffected by outcome feedback about whether an outbreak had occurred in the previous round but were strongly influenced by social feedback about other group members' prior biosecurity decisions: the probability of biosecuring increased as a function of the proportion of other players that chose to biosecure in the previous round. Critically, the effect of social feedback was considerably stronger in the presence than in the absence of communication. This result is

consistent with the notion many players are conditional cooperators who are willing to cooperate if they think that others will cooperate as well (Fischbacher & Gächter, 2010b; Fischbacher et al., 2001; Isler et al., 2021; Keser & Van Winden, 2000; Kocher et al., 2008; Kopsacheilis et al., 2025; Kurzban & Houser, 2005; Thöni & Volk, 2018) and that communication provided a mechanism for such individuals to signal one another and reinforce conditional cooperation.

This behavioural evidence converged with the results of the player type, trust and self-reported strategy use, and communication analyses. Starting with the analysis of player types based on biosecurity decision profiles under high risk, in the absence of communication, potential free riders represented the largest group, whereas in the presence of communication the majority of players were classified as conditional cooperators and none were classified as potential free riders. Participants in the communication treatment also reported greater trust in their group members, and were more likely to self-identify as conditional cooperators and less likely to self-identify as non-contributors, compared to participants in the no-communication treatment. Finally, consistent with previous studies of communication in the public goods game and other social dilemmas, the communication analyses revealed that communication was largely used as a coordination device (Bochet et al., 2006; Brosig et al., 2003; Isaac & Walker, 1988; Ostrom & Walker, 1997; Ostrom et al., 1994): groups used communication to negotiate strategies tailored towards the risk level, to exchange commitments to cooperate, and to agree what each member should do in subsequent rounds.

Turning to the association between biosecurity decisions and risk and loss aversion, in the absence of communication, risk aversion was positively associated with biosecurity decisions: as risk aversion increased, so too did the probability of biosecuring. However, this association was abolished in the presence of communication. This pattern was observed for both the self-report and the parameter-based measures of risk aversion. This result suggests that when communication was available, biosecurity decisions were guided

by group-level strategies rather than individual risk preferences. By contrast, there was no reliable association between loss aversion and biosecurity decisions, for either the self-report or parameter-based measures, despite loss aversion being the strategy most frequently selected by participants in the post-game questionnaire. This pattern suggests that although participants consciously perceived loss aversion as central to their decision-making, it did not translate into a consistent pattern of behavioural responses.

6.2 Implications for theory

The results are broadly consistent with the Hennessy (2008) model of biosecurity. Under the entry-risk parameter settings examined in our experiment, the incentive to biosecure is greater under high risk than under low risk, and the behavioural responses of our groups of participants were sensitive to these differences in the underlying incentive structures. However, according to the model, in both risk scenarios, mutual non-biosecuring is the dominant strategy and the unique Nash equilibrium, yet in all treatments participants biosecured at rates substantially greater than predicted by the theory. This result is not surprising and is consistent with experimental evidence from the prisoner's dilemma (Kiyonari et al., 2000; Sally, 1995) and public goods games (Chaudhuri, 2011; Ledyard, 1995) showing that cooperation occurs at a rate much greater than predicted by standard economic theory. The findings are therefore consistent with the model's assumptions about how the entry risk alters the relative incentive to biosecure, while at the same time demonstrating that observed behaviour cannot be fully explained in terms of standard assumptions of rationality and self-interest.

Hennessy's one-shot model assumes that in choosing how to respond, actors form beliefs about how many others will biosecure. However, it does not specify how these beliefs are formed through repeated interaction or how they are influenced by communication and feedback about others' past actions. Our experimental findings identify conditional cooperation as one potential mechanism by which these expectations are established, especially when opportunities for communication are available. In our

experiment, communication allowed players to exchange commitments to cooperate, while subsequent social feedback allowed them to verify whether those commitments had been honoured. This combination appears to have increased trust and strengthened the tendency of individuals to reciprocate the biosecurity decisions of others.

It is possible that a minimally augmented version of the Hennessy model can explain some of the evidence suggestive of conditional cooperation. For example, in the online resources (Supplementary Game-Theoretic Analysis), we present a repeated version of the model that takes into account loss aversion and risk aversion. We show that if a sufficient number of players are either loss averse or risk averse, then this converts the prisoner's dilemma under high risk into a coordination game, generating predicted outcomes that resemble conditional cooperation. The problem for this account is that we find no link between loss aversion and biosecurity decisions in our experiment, and whilst risk aversion was a positive predictor of biosecurity decisions in the absence of communication it was not a reliable predictor in the presence of communication—that is, under the very conditions where the evidence of conditionally cooperative behaviour is strongest.

It seems likely that an adequate account of these findings will require a behavioural extension of the model that incorporates the notion of social preferences. This would require augmenting the payoff function to include not only a monetary utility component but a psychological utility component, perhaps corresponding to reciprocity or guilt aversion, that increases as a function of the number of actors that biosecured in the previous round and is weighted by a parameter reflecting the attention given to this social component. This social component would thereby capture the tendency for individuals to derive greater psychological utility from biosecuring when others also biosecure and would provide a mechanism whereby the temptation to free ride on the biosecurity actions of others can be offset by the psychological desire to reciprocate their actions. The stronger evidence of conditional cooperation in the presence of communication could be explained by an increase in the attentional weight placed on this social component based on the

assumption that opportunities for social exchange make reciprocal expectations and the social consequences of violating them more salient.

Consistent with research using the public goods game and the strategy method (Fischbacher & Gächter, 2010b; Fischbacher et al., 2001; Kocher et al., 2008; Thöni & Volk, 2018), we find evidence for the existence of different player types with different cooperative tendencies (see also Ostrom, 2000). However, as previously noted, there are limitations to our approach to classifying players, which was based on their responses to the range of decisions of other group members that occurred in the game, rather than eliciting their responses to a range of possible decisions of other group members, as is the case using the strategy method. This means that the classification of a player may be dependent upon the conditions that they encountered in the experiment and may not reflect their underlying cooperative dispositional tendencies. For this reason, we think it would be premature to speculate on how the Hennessy model could be modified to accommodate the evidence of different player types.

6.3 Implications for policy and practice

The first implication of our findings concerns the sensitivity of biosecurity collective action to the risk level and individual risk aversion. Participants biosecured more frequently when the risk of an outbreak was high than when it was low, indicating that they were able to condition their degree of biosecurity effort on the risk environment. In the absence of communication, greater individual risk aversion was reliably associated with a higher likelihood of biosecuring. In the experiment, participants were given precise information about the risk of an outbreak occurring and how it varied as a function of the number of individuals choosing to biosecure. However, despite the availability of complete and accurate information, biosecurity was underprovided when the risk was high and overprovided when the risk was low.

In real-world settings, locally relevant and credible information about the risk level would need to be communicated to farmers by appropriate biosecurity organisations and

this information will necessarily be less precise than under the idealised conditions of our experiment. Without clear and credible information, farmers may misperceive the objective risk of an outbreak occurring: if they perceive the risk to be lower than the objective risk level, then they may expend insufficient biosecurity effort, potentially resulting in costly outbreaks; conversely, if they perceive the risk to be higher than the objective risk level, then they may expend more biosecurity effort than is warranted, potentially resulting in wasted expenditure.

Yet the manner in which risk information is expressed may be just as important as its accuracy given the well-documented difficulties people encounter interpreting and reasoning about probabilities. For example, it is well known that people tend to overweight small probabilities and underweight moderate and large probabilities (Gonzalez & Wu, 1999; Kahneman & Tversky, 1979; Prelec, 1998; Tversky & Kahneman, 1992; Wu & Gonzalez, 1996). The latter finding may partially explain the underprovision of biosecurity in the high-risk treatment of our experiment. One potential solution is to communicate numerical information using frequency formats, as several lines of evidence suggest that people are better at comprehending numerical information when it is expressed as frequencies (e.g., 59 out of 100) than when the same information is expressed as percentages (e.g., 59%) or probabilities (e.g., 0.59; Gigerenzer & Hoffrage, 1995; Newell et al., 2008; Slovic et al., 2000; K. Yamagishi, 1997). Other research on individual judgement and uncertainty communication suggests there may be additional merit in combining numerical information with verbal statements of likelihood (e.g., "likely" or "very likely") to minimise problems associated with the misinterpretation of numerical information (Budescu et al., 2009, 2012, 2014). Biosecurity organisations might, for example, communicate the expected number of outbreaks over 100 seasons based on the risk level and differing degrees of biosecurity participation and effort, accompanied by an appropriate verbal statement of the likelihood of an outbreak occurring.

A related conceptual challenge is communicating the joint nature of pest freedom.

Specifically, that maintaining pest-free status depends not on individual action alone, but on the collective action of all farmers. In the experiment, participants were informed about this collective-risk dimension of biosecurity and their behaviour reflected this understanding of the decision problem. However, to our knowledge this is not a concept that is typically communicated by biosecurity organisations. In Australia, for example, biosecurity governance arrangements emphasise the role of individual risk management and responsibility among biosecurity stakeholders, rather than coordinated collective action (Maru et al., 2025), although there are important exceptions in the form of area-wide management schemes that we consider below. The recognition of biosecurity as a social dilemma with public goods characteristics requires a shift from an individual-focused to a collaborative-focused approach. Achieving this transformation may require coordinating organisations to explicitly highlight the collective-risk nature of biosecurity. A focus on communicating how the probability of an outbreak affecting an entire region depends not merely on the entry risk but also on how the actions of multiple actors combine to increase or reduce the probability, would underscore how potentially damaging free riding is not just in a single production season but over many seasons if export markets are lost.

Our findings concerning communication also have implications for communication between biosecurity stakeholders engaged in collective pest and disease management schemes. In Australia, specific examples of long-standing collective pest or area-wide management schemes for fruit fly include the Central Burnett QFly zone (Lloyd et al., 2010) and Vinehealth Australia (2025). These schemes are run by highly sophisticated industry organisations that manage communication with industry and the general public, as well as technical requirements for border protection and eradication (Kruger, 2017, 2021). These schemes engage in intensive communication about best practice, the extent of pest incursions, and regional border protection activities. Communication is directed towards both commercial farmers, to advise them about incursions and export restrictions, and the general public, especially gardeners, whose actions can also influence the risk of

pest incursions.

Linking this study to those schemes, the communication simulated in the experiment was analogous to that between commercial farmers. It had the effect of increasing trust and contributions to collective biosecurity. Communication during pest outbreaks and at other times tends to be formal and top-down from the coordinating organisation to farmers. Our study indicates that there may also be a critical role for communication between farmers, possibly moderated by the coordinator. For example, the coordinator could facilitate regular meetings at which participants consider the prevailing risk, agree upon an appropriate collective response, and clarify their respective responsibilities.

That communication between farmers may facilitate cooperation is consistent with observational evidence showing that a defining feature of the more effective area-wide management schemes for QFly is strong two-way communication, including frequent face-to-face interactions between farmers and other stakeholders (e.g., crop consultants, researchers, and members of the public) to promote trust. Regions with strong communication networks are characterised by high levels of participation and cooperation, whereas regions with weak networks are characterised by low levels of participation and greater free riding (Kruger, 2016a, 2016b). Although observational studies cannot confirm causality, our controlled experiment provides causal evidence that communication can increase cooperation in a collective biosecurity setting. Together, these findings suggest that facilitating communication between farmers should be considered an important component of collective pest management schemes.

One important consideration is that the observed benefits of communication in our experiment occurred in the presence of social feedback about the biosecurity decisions of each group member. Biosecurity decisions were sensitive to this information: the probability of biosecuring increased as a function of the proportion of other players that biosecured on the previous round, and this effect was especially pronounced when communication was available. As previously noted, this is presumably because social

feedback afforded players the opportunity to verify whether their group members had honoured their commitments. The important thing to note is that in the experiment this social feedback was provided automatically. However, comparable information would not necessarily arise naturally in real-world biosecurity settings, where measures undertaken on private properties may be difficult for others to observe. Instead, regular monitoring of the compliance of farmers with agreed responsibilities may be necessary to ensure that everyone is doing their part. Ideally, this monitoring should be undertaken by the farmers themselves and other local stakeholders as this helps promote trust and foster mutual accountability (Ostrom et al., 1990). Importantly, monitoring does not come for free as it did in our experiment, since it requires management time and other resources.

These conclusions must, however, be qualified by an important caveat. Our results do not show that monitoring itself increases cooperation because social feedback was present in all treatments and we lack the counterfactual scenario in which social feedback was absent. What our results do show, however, is that communication makes information about others' behaviour more consequential when it is available.

6.4 Potential limitations and future directions

Before concluding, there are some potential limitations of our study that merit comment. First, our entry risk manipulation was confined to a comparison between two of the three strategic regions identified within the Hennessy (2008) model. Specifically, we compared a high-risk treatment, representing the prisoner's dilemma region, with a low-risk treatment, representing the no-dilemma region. The prisoner's dilemma region was of key interest because it is where biosecurity is most vulnerable to being underprovided, whereas the no-dilemma region served as a baseline comparison scenario characterised by the absence of a social dilemma. However, constraints on the length of experimental sessions meant that we did not include a very high-risk treatment corresponding to the coordination game region. The provision of biosecurity should be easier within this region because mutual biosecuring is a Nash equilibrium, and therefore the benefit of

communication may be smaller. This is a theoretical prediction that will need to be tested in future experimental work using the biosecurity collective-risk social dilemma.

Second, following Hennessy (2008), we adopted the simplifying assumption that full cooperation eliminates the risk of pest entry. However, this is probably not a plausible assumption for most invasive species. In most cases, even if all parties biosecured, there would still be some residual probability of invasion occurring. The fact that the risk of pest entry could be eradicated in our experiment may have influenced biosecurity decisions. If some residual probability of pest entry remained even under the best-case scenario of full cooperation, this might render individuals more reluctant to engage in risk-reducing action. This is a possibility that warrants exploring in future extensions of the current work.

Finally, another limitation of our current game is the binary nature of biosecurity decisions—players must choose either to biosecure or not. This contrasts with real-world biosecurity practices, where farmers can adjust their efforts depending on the risk level. Notably, some group strategies observed in the game—such as alternating cooperation in pairs across rounds—can be seen as an attempt to approximate partial effort. For example, when the entry risk is low, framers might agree to spray only a portion of their crops. Our decision to simulate biosecurity in binary terms was intentional, aligning with Hennessy's (2008) model, which also treats biosecurity as an all-or-nothing choice. Nonetheless, extending the model and game to incorporate a continuum of effort would enhance their ecological validity and permit the observation of more nuanced strategic behaviour.

In this experiment, we examined the impact of risk and communication on collective biosecurity decision making. However, the game can readily be adapted to model the impact of other key variables. Hennessy's (2008) model, for example, makes specific predictions regarding the impact of heterogeneity in biosecurity costs and values at risk, the role of leadership, and the effect of large group interactions—where behaviour is expected to depart from that observed in smaller groups. Future work could extend the game to test these variables.

6.5 Conclusions

We introduced a novel economic game—the biosecurity collective-risk social dilemma—for simulating the collective action problem involved in biosecurity. Consistent with the predictions of Hennessy's (2008) model on which the game is based, participants were more likely to biosecure when the risk of pest entry was high than when it was low, although biosecurity remained underprovided in the high-risk treatment and was overprovided in the low-risk treatment. The opportunity to communicate on certain rounds increased engagement in risk-reducing action by increasing trust, providing opportunities for conditional cooperation, and facilitating strategic coordination. The results highlight the importance of effective risk communication and opportunities for communication among biosecurity stakeholders in real-world pest management schemes and suggest new opportunities for strengthening biosecurity protections. The biosecurity collective-risk social dilemma provides a flexible experimental framework for investigating other factors that may promote or inhibit collective biosecurity and examining their implications for policy and practice.

Declaration of competing interest

All authors declare no other financial or non-financial competing interests.

Data availability

Data are available online at:
https://anonymous.4open.science/r/biosec-supplement-2025.

Code availability

Analysis scripts are available online at:
https://anonymous.4open.science/r/biosec-supplement-2025.

References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Trans. Autom. Control*, *19*, 716–723.

Athanassiou, C. G., Phillips, T. W., & Wakil, W. (2019). Biology and control of the khapra beetle, trogoderma granarium, a major quarantine threat to global food security. *Annu. Rev. Entomol.*, *64*, 131–148.

Bagavathiannan, M. V., Graham, S., Ma, Z., Barney, J. N., Coutts, S. R., Caicedo, A. L., Clerck-Floate, D., West, N. M., Blank, L., Metcalf, A. L., et al. (2019). Considering weed management as a social dilemma bridges individual and collective interests. *Nat. Plants*, *5*(4), 343–351.

Bochet, O., Page, T., & Putterman, L. (2006). Communication and punishment in voluntary contribution experiments. *J. Econ. Behav. Organ.*, *60*(1), 11–26.

Brosig, J., Weimann, J., & Ockenfels, A. (2003). The effect of communication media on cooperation. *Ger. Econ. Rev.*, *4*(2), 217–241.

Budescu, D. V., Broomell, S., & Por, H.-H. (2009). Improving communication of uncertainty in the reports of the intergovernmental panel on climate change. *Psychol. Sci.*, *20*(3), 299–308.

Budescu, D. V., Por, H.-H., & Broomell, S. B. (2012). Effective communication of uncertainty in the ipcc reports. *Clim. Change*, *113*(2), 181–200.

Budescu, D. V., Por, H.-H., Broomell, S. B., & Smithson, M. (2014). The interpretation of ipcc probabilistic statements around the world. *Nat. Clim. Change*, *4*(6), 508–512.

Burnett, K. M. (2006). Introductions of invasive species: Failure of the weaker link. *Agric. Resour. Econ. Rev.*, *35*(1), 21–28.

Camerer, C. F. (2003). *Behavioural game theory*. Princeton: Princeton University Press.

Carnegie, A. (2015). First report of puccinia psidii (myrtle rust) in eucalyptus plantations in australia. *Plant Dis.*, *99*(1), 161–161.

Chaudhuri, A. (2011). Sustaining cooperation in laboratory public goods experiments: A selective survey of the literature. *Exp. Econ.*, *14*(1), 47–83.

Cobourn, K. M., Amacher, G. S., & Haight, R. G. (2019). Cooperative management of invasive species: A dynamic nash bargaining approach. *Environ. Resour. Econ.*, *72*(4), 1041–1068.

Dawes, R. M., McTavish, J., & Shaklee, H. (1977). Behavior, communication, and assumptions about other people's behavior in a commons dilemma situation. *J. Pers. Soc. Psychol.*, *35*(1), 1–11.

De Lima, C. P. F., Poogoda, S. R., Mansfield, E. R., & Collie, H. M. (2025). Suppression of the mediterranean fruit fly in semi-temperate areas of western australia: A review of over 30 years of area wide management practice. *J. Appl. Entomol.*

Dominiak, B. C., & Ekman, J. H. (2013). The rise and demise of control options for fruit fly in australia. *Crop Prot.*, *51*, 57–67.

Donaldson, A. (2008). Biosecurity after the event: Risk politics and animal disease. *Environ. Plann. A*, *40*(7), 1552–1567.

Fehr, E., & Gächter, S. (2002). Altruistic punishment in humans. *Nature*, *415*(6868), 137–140.

Fehr, E., & Schmidt, K. M. (2006). The economics of fairness, reciprocity and altruism–experimental evidence and new theories. In S.-C. Kolm & Y. J. M. (Eds.), *Handbook of the economics of giving, altruism and reciprocity* (pp. 615–691, Vol. 1). Amsterdam: Elsevier.

Fischbacher, U. (2007). Z-tree: Zurich toolbox for ready-made economic experiments. *Exp. Econ.*, *10*(2), 171–178.

Fischbacher, U., & Gächter, S. (2010a). Social preferences, beliefs, and the dynamics of free riding in public goods experiments. *Am. Econ. Rev.*, *100*(1), 541–56.

Fischbacher, U., & Gächter, S. (2010b). Social preferences, beliefs, and the dynamics of free riding in public goods experiments. *Am. Econ. Rev.*, *100*(1), 541–556.

1207 Fischbacher, U., Gächter, S., & Fehr, E. (2001). Are people conditionally cooperative?
1208 evidence from a public goods experiment. *Econ. Lett.*, *71*(3), 397–404.
1209 Florec, V., Sadler, R. J., White, B., & Dominiak, B. C. (2013). Choosing the battles: The
1210 economics of area wide pest management for queensland fruit fly. *Food Policy*, *38*,
1211 203–213.
1212 Gächter, S., & Fehr, E. (1999). Collective action as a social exchange. *J. Econ. Behav.*
1213 *Organ.*, *39*(4), 341–369.
1214 Gächter, S., & Herrmann, B. (2006). Human cooperation from an economic perspective. In
1215 P. M. Kappeler & C. P. van Schaik (Eds.), *Cooperation in primates and humans:*
1216 *Mechanisms and evolution* (pp. 279–301). Springer.
1217 Gachter, S., Renner, E., & Sefton, M. (2008). The long-run benefits of punishment.
1218 *Science*, *322*(5907), 1510–1510.
1219 Garcia-Figuera, S., Grafton-Cardwell, E. E., Babcock, B. A., Lubell, M. N., &
1220 McRoberts, N. (2021). Institutional approaches for plant health provision as a
1221 collective action problem. *Food Secur.*, *13*(2), 273–290.
1222 Gigerenzer, G., & Hoffrage, U. (1995). How to improve bayesian reasoning without
1223 instruction: Frequency formats. *Psychol. Rev.*, *102*(4), 684–704.
1224 Gonzalez, R., & Wu, G. (1999). On the shape of the probability weighting function. *Cogn.*
1225 *Psychol.*, *38*(1), 129–166.
1226 Greiner, B. (2015). Subject pool recruitment procedures: Organizing experiments with
1227 orsee. *J. Eur. Econ. Assoc.*, *1*(1), 114–125.
1228 Hennessy, D. A. (2007). Biosecurity and spread of an infectious animal disease. *Am. J.*
1229 *Agric. Econ.*, *89*(5), 1226–1231.
1230 Hennessy, D. A. (2008). Biosecurity incentives, network effects, and entry of a rapidly
1231 spreading pest. *Ecol. Econ.*, *68*(1-2), 230–239.
1232 Isaac, R. M., Schmidtz, D., & Walker, J. M. (1989). The assurance problem in a laboratory
1233 market. *Public Choice*, *62*(3), 217–236.

Isaac, R. M., & Walker, J. M. (1988). Communication and free-riding behavior: The voluntary contribution mechanism. *Econ. Inq.*, *26*(4), 585–608.

Isler, O., Gächter, S., Maule, A. J., & Starmer, C. (2021). Contextualised strong reciprocity explains selfless cooperation despite selfish intuitions and weak social heuristics. *Sci. Rep.*, *11*(1), 13868.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, *47*, 263–291.

Keser, C., & Van Winden, F. (2000). Conditional cooperation and voluntary contributions to public goods. *Scand. J. Econ.*, *102*(1), 23–39.

Kiyonari, T., Tanida, S., & Yamagishi, T. (2000). Social exchange and reciprocity: Confusion or a heuristic? *Evol. Hum. Behav.*, *21*(6), 411–427.

Kocher, M. G., Cherry, T., Kroll, S., Netzer, R. J., & Sutter, M. (2008). Conditional cooperation on three continents. *Econ. Lett.*, *101*(3), 175–178.

Kopsacheilis, O., van Dolder, D., & Isler, O. (2025). Conditional cooperation under uncertainty: The social description-experience gap. *Games Econ. Behav.*, *152*, 241–256.

Kruger, H. (2016a). Adaptive co-management for collaborative commercial pest management: The case of industry-driven fruit fly area-wide management. *Int. J. Pest Manag.*, *62*(4), 336–347.

Kruger, H. (2016b). Designing local institutions for cooperative pest management to underpin market access: The case of industry-driven fruit fly area-wide management. *Int. J. Commons*, *10*(1), 176–199.

Kruger, H. (2017). Creating an enabling environment for industry-driven pest suppression: The case of suppressing queensland fruit fly through area-wide management. *Agric. Syst.*, *156*, 139–148.

Kruger, H. (2021). Industry-driven area-wide management of queensland fruit fly in queensland and new south wales, australia: Can it work? In *Area-wide integrated pest management* (pp. 693–708). CRC Press.

Kurzban, R., & Houser, D. (2005). Experiments investigating cooperative types in human groups: A complement to evolutionary theory and simulations. *Proc. Natl. Acad. Sci. U.S.A.*, *102*(5), 1803–1807.

Lampert, A. (2020). Multiple agents managing a harmful species population should either work together to control it or split their duties to eradicate it. *Proc. Natl. Acad. Sci. U.S.A.*, *117*(19), 10210–10217.

Ledyard, J. O. (1995). Public goods: A survey of experimental research. In J. Kagel & A. Roth (Eds.), *Handbook of experimental economics*. Princeton: Princeton University Press.

Lloyd, A. C., Hamacek, E. L., Kopittke, R. A., Peek, T., Wyatt, P. M., Neale, C. J., Eelkema, M., & Gu, H. (2010). Area-wide management of fruit flies (diptera: Tephritidae) in the central burnett district of queensland, australia. *Crop Prot.*, *29*(5), 462–469.

Maru, Y. T., Kruger, H., Loechel, B., Hernandez-Jover, M., Kelly, J., Manyweathers, J., & El Hassan, M. (2025). Introducing institutional design principles for transforming on-ground biosecurity. *Agric. Syst.*, *229*, 104402.

Masclet, D., Noussair, C., Tucker, S., & Villeval, M.-C. (2003). Monetary and nonmonetary punishment in the voluntary contributions mechanism. *Am. Econ. Rev.*, *93*(1), 366–380.

Matuschek, H., Kliegl, R., Vasishth, S., Baayen, H., & Bates, D. (2017). Balancing Type I error and power in linear mixed models. *J. Mem. Lang.*, *94*, 305–315.

Milinski, M., Semmann, D., & Krambeck, H.-J. (2002). Reputation helps solve the 'tragedy of the commons'. *Nature*, *415*(6870), 424–426.

Milinski, M., Sommerfeld, R. D., Krambeck, H.-J., Reed, F. A., & Marotzke, J. (2008). The collective-risk social dilemma and the prevention of simulated dangerous climate change. *Proc. Natl. Acad. Sci. U.S.A.*, *105*(7), 2291–2294.

Mohammad, S., & Turney, P. (2010). Emotions evoked by common words and phrases: Using mechanical turk to create an emotion lexicon. *Proceedings of the NAACL HLT 2010 workshop on computational approaches to analysis and generation of emotion in text*, 26–34.

Newell, B. R., Mitchell, C. J., & Hayes, B. K. (2008). Getting scarred and winning lotteries: Effects of exemplar cuing and statistical format on imagining low-probability events. *J. Behav. Decis. Mak.*, *21*(3), 317–335.

Nielsen, F. (2011). Evaluation of a word list for sentiment analysis in microblogs. arxiv 2011. *arXiv preprint arXiv:1103.2903*.

Olson, L. J., & Roy, S. (2002). The economics of controlling a stochastic biological invasion. *Am. J. Agric. Econ.*, *84*(5), 1311–1316.

Olson, L. J., & Roy, S. (2008). Controlling a biological invasion: A non-classical dynamic economic model. *Econ. Theory*, *36*(3), 453–469.

Ostrom, E., et al. (1990). *Governing the commons: The evolution of institutions for collective action*. Cambridge: Cambridge University Press.

Ostrom, E. (2000). Collective action and the evolution of social norms. *J. Econ. Perspect.*, *14*(3), 137–158.

Ostrom, E. (2003). Toward a behavioral theory linking trust, reciprocity, and reputation. In E. Ostrom & J. Walker (Eds.), *Trust and reciprocity: Interdisciplinary lessons from experimental research* (pp. 19–79). Russell Sage Foundation: New York.

Ostrom, E., Gardner, R., & Walker, J. (1994). *Rules, games, and common-pool resources*. University of Michigan press.

Ostrom, E., & Walker, J. (1997). Neither markets nor states: Linking transformation processes in collective action arenas. In D. C. Mueller (Ed.), *Perspectives on public choice: A handbook.* Cambridge University Press.

Page, T., Putterman, L., & Unel, B. (2005). Voluntary association in public goods experiments: Reciprocity, mimicry and efficiency. *Econ. J.*, *115*(506), 1032–1053.

Perrings, C. (2016). Options for managing the infectious animal and plant disease risks of international trade. *Food Secur.*, *8*(1), 27–35.

Perrings, C., Williamson, M., Barbier, E. B., Delfino, D., Dalmazzone, S., Shogren, J., Simmons, P., & Watkinson, A. (2002). Biological invasion risks and the public good: An economic perspective. *Conserv. Ecol.*, *6*(1).

Pisor, A. C., Gervais, M. M., Purzycki, B. G., & Ross, C. T. (2020). Preferences and constraints: The value of economic games for studying human behaviour. *R. Soc. Open Sci.*, *7*(6), 192090.

Prelec, D. (1998). The probability weighting function. *Econometrica*, *66*(3), 497–527.

R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria.

Rege, M., & Telle, K. (2004). The impact of social approval and framing on cooperation in public good situations. *J. Public Econ.*, *88*(7-8), 1625–1644.

Sally, D. (1995). Conversation and cooperation in social dilemmas: A meta-analysis of experiments from 1958 to 1992. *Ration. Soc.*, *7*(1), 58–92.

Schelling, T. C. (1980). *The strategy of conflict*. Harvard University Press.

Schwartz, G. (1978). Estimating the dimension of a model. *Ann. Stat.*, *6*, 461–464.

Semmann, D., Krambeck, H.-J., & Milinski, M. (2004). Strategic investment in reputation. *Behav. Ecol. Sociobiol.*, *56*(3), 248–252.

Slovic, P., Monahan, J., & MacGregor, D. G. (2000). Violence risk assessment and risk communication: The effects of using actual cases, providing instruction, and employing probability versus frequency formats. *Law Hum. Behav.*, *24*(3), 271–296.

Suleiman, R., & Rapoport, A. (1988). Environmental and social uncertainty in single-trial resource dilemmas. *Acta Psychol.*, *68*(1-3), 99–112.

Sultana, S., Baumgartner, J. B., Dominiak, B. C., Royer, J. E., & Beaumont, L. J. (2020). Impacts of climate change on high priority fruit fly species in australia. *PLoS One*, *15*(2), e0213820.

Tam, M., Capon, T., Whitten, S., Tapsuwan, S., Kandulu, J., & Measham, P. (2023). Assessing the economic benefit of area wide management and the sterile insect technique for the queensland fruit fly in pest-free vs. endemic regions of south-east australia. *Int. J. Pest Manag.*, *69*(1), 64–80.

Tanaka, T., Camerer, C. F., & Nguyen, Q. (2010). Risk and time preferences: Linking experimental and household survey data from vietnam. *Am. Econ. Rev.*, *100*(1), 3–25.

Tapsuwan, S., Capon, T., Tam, M., Kandulu, J., Measham, P., & Whitten, S. (2020). Willingness to pay for area-wide management and sterile insect technique to control fruit flies in australia. *Int. J. Pest Manag.*, *66*(4), 351–367.

Thielmann, I., Böhm, R., Ott, M., & Hilbig, B. E. (2021). Economic games: An introduction and guide for research. *Collabra Psychol.*, *7*(1), 19004.

Thöni, C., & Volk, S. (2018). Conditional cooperation: Review and refinement. *Econ. Lett.*, *171*, 37–40.

Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *J. Risk Uncertain.*, *5*(4), 297–323.

Wu, G., & Gonzalez, R. (1996). Curvature of the probability weighting function. *Manag. Sci.*, *42*(12), 1676–1690.

Wylie, F. R., & Janssen-May, S. (2017). Red imported fire ant in australia: What if we lose the war? *Ecol. Manag. Restor.*, *18*(1), 32–44.

Yamagishi, K. (1997). When a 12.86% mortality is more dangerous than 24.14%: Implications for risk communication. *Appl. Cogn. Psychol.*, *11*(6), 495–506.

1364 Yamagishi, T., & Yamagishi, M. (1994). Trust and commitment in the united states and
1365 japan. *Motiv. Emot.*, *18*(2), 129–166.
1366 Zelmer, J. (2003). Linear public goods experiments: A meta-analysis. *Exp. Econ.*, *6*(3),
1367 299–310.

Table 1

Calculation of expected payoffs and protection probabilities for each combination of firm strategies.

		Firm 2 strategy	
Firm 1 strategy	Outcome	Biosecure	Not biosecure
Biosecure	Expected payoff	$\{\pi_1 = v_1 - c_1^b - c_1^p,\ \pi_2 = v_2 - c_2^b - c_2^p\}$	$\{\pi_1 = \sigma v_1 - c_1^b - c_1^p,\ \pi_2 = \sigma v_2 - c_2^p\}$
	Protection probability	1	σ
Not biosecure	Expected payoff	$\{\pi_1 = \sigma v_1 - c_1^p,\ \pi_2 = \sigma v_2 - c_2^b - c_2^p\}$	$\{\pi_1 = \sigma^2 v_1 - c_1^p,\ \pi_2 = \sigma^2 v_2 - c_2^p\}$
	Protection probability	σ	σ^2

Table 2

Probability of an outbreak of pests as a function of risk treatment and number of players choosing to protect.

	Number of players choosing to:		
Risk treatment	Protect	Not protect	Probability of an outbreak
Low risk	0	4	0.34
	1	3	0.27
	2	2	0.19
	3	1	0.10
	4	0	0
High risk	0	4	0.59
	1	3	0.49
	2	2	0.36
	3	1	0.20
	4	0	0

Table 3

Net payoffs on each round as a function of whether a player chooses to protect and whether an outbreak of pests occurs.

	Net payoff	
Player decision	No outbreak	Outbreak
Protect	$v - c^b - c^p = \$25 - \$10 - \$5 = \$10$	$v - c^b - c^p = \$0 - \$10 - \$5 = -\$15$
Not protect	$v - c^p = \$25 - \$5 = \$20$	$v - c^p = \$0 - \$5 = -\$5$

Table 4

Estimated generalized linear mixed-effects model parameters for the treatment-only model predicting biosecurity decisions.

Fixed Effects					
Effect	β	SE	95% CI	z	p
Intercept	−2.50	0.50	[−3.47, −1.52]	−5.02	< .001
Risk	1.88	0.28	[1.33, 2.42]	6.76	< .001
Communication	2.22	0.69	[0.87, 3.56]	3.22	.001
Round	−0.62	0.28	[−1.17, −0.08]	−2.24	.025
Risk × Communication	0.78	0.38	[0.03, 1.53]	2.05	.041
Risk × Round	−0.22	0.17	[−0.56, 0.12]	−1.26	.207
Communication × Round	0.50	0.38	[−0.25, 1.25]	1.31	.191
Risk × Communication × Round	0.06	0.24	[−0.41, 0.52]	0.24	.808

Random Effects			
Level	Effect	Variance	SD
Player	Intercept	2.20	1.48
	Risk	1.83	1.35
	Round	0.28	0.53
Group	Intercept	2.04	1.43
	Round	0.62	0.79

Model Fit		
R^2	Marginal	Conditional
	0.29	0.71

Note—p-values for fixed effects were calculated using asymptotic Wald tests. Confidence intervals were calculated using the Wald method. The model included by-player random slopes for risk and round, and a by-group random slope for round. Random-effect correlations were $r = -0.68$ for player intercept–risk, $r = 0.79$ for player intercept–round, $r = -0.70$ for player risk–round, and $r = 0.31$ for group intercept–round.

Table 5

Goodness-of-fit quantities for the four generalized linear mixed-effects models predicting biosecurity decisions.

			Model fit					LRT vs. nested		
Model name	Nested model	k	$\ln L$	AIC	ΔAIC	BIC	ΔBIC	df	χ^2	p
Treatment only		17	−1166.47	2366.94	94.58	2467.18	71.00			
Additive feedback	Treatment only	19	−1125.18	2288.36	16.01	2400.40	4.22	2	82.57	$< .001$
Social feedback	Additive feedback	21	−1115.18	2272.35	0.00	2396.18	0.00	2	20.01	$< .001$
Outcome feedback	Additive feedback	21	−1123.28	2288.57	16.22	2412.39	16.22	2	3.80	.150

Note—$\ln L$ = natural logarithm of the maximum likelihood; k = number of estimated parameters; AIC = Akaike Information Criterion; ΔAIC = difference in AIC with respect to the best-fitting model; BIC = Bayesian Information Criterion; ΔBIC = difference in BIC with respect to the best-fitting model; df, χ^2, and p = degrees of freedom, chi-square, and p value, respectively, for Likelihood ratio test (LRT). For the LRT, a significant p value shows that the model indicated by the row provides a significantly better fit than its nested counterpart.

Table 6

Estimated generalized linear mixed-effects model parameters for the social-feedback model predicting biosecurity decisions.

Fixed Effects					
Effect	β	SE	95% CI	z	p
Intercept	−2.66	0.44	[−3.52, −1.79]	−6.02	< .001
Risk	1.19	0.34	[0.52, 1.86]	3.47	< .001
Communication	0.88	0.60	[−0.28, 2.05]	1.48	.138
Round	−0.64	0.20	[−1.02, −0.26]	−3.27	.001
Lagged Outcome	0.17	0.14	[−0.10, 0.44]	1.26	.207
Lagged Protection	0.71	0.54	[−0.34, 1.77]	1.32	.185
Risk × Communication	−0.42	0.42	[−1.25, 0.42]	−0.98	.327
Risk × Round	−0.02	0.19	[−0.40, 0.35]	−0.12	.901
Communication × Round	0.47	0.25	[−0.03, 0.96]	1.85	.064
Risk × Lagged Protection	0.86	0.50	[−0.12, 1.85]	1.72	.085
Communication × Lagged Protection	2.47	0.56	[1.37, 3.56]	4.40	< .001
Risk × Communication × Round	−0.04	0.26	[−0.55, 0.47]	−0.15	.878

Random Effects			
Level	Effect	Variance	SD
Player	Intercept	2.28	1.51
	Risk	1.41	1.19
	Round	0.36	0.60
Group	Intercept	1.07	1.03
	Round	0.06	0.25

Model Fit		
R^2	Marginal	Conditional
	0.36	0.70

Note—p-values for fixed effects were calculated using asymptotic Wald tests. Confidence intervals were calculated using the Wald method. The model included by-player random slopes for risk and round, and a by-group random slope for round. Random-effect correlations were $r = -0.47$ for player intercept–risk, $r = 0.65$ for player intercept–round, $r = -0.39$ for player risk–round, and $r = 0.35$ for group intercept–round.

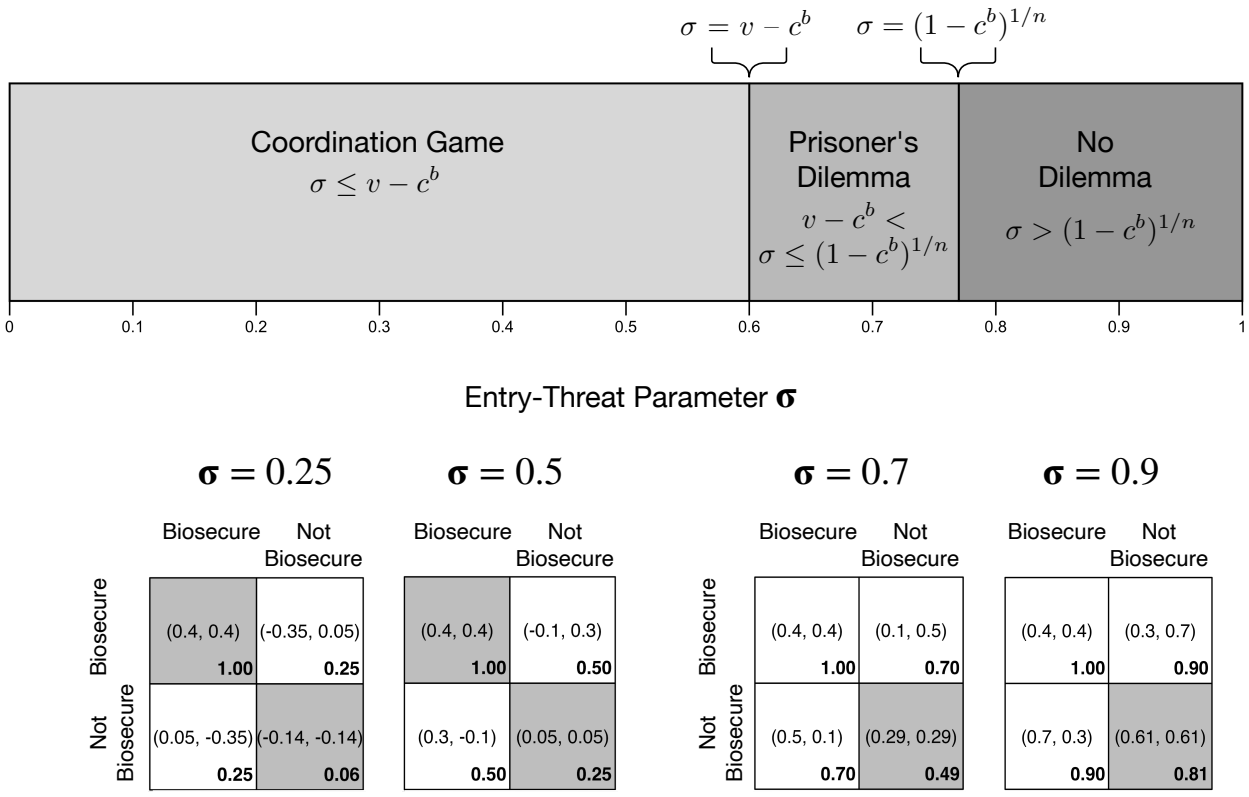


Figure 1

The three regions of the entry-threat parameter space, each illustrated with one or more payoff matrices. In the matrices, bold-face values represent protection probabilities, whereas shaded cells represent pure strategy Nash equilibria.

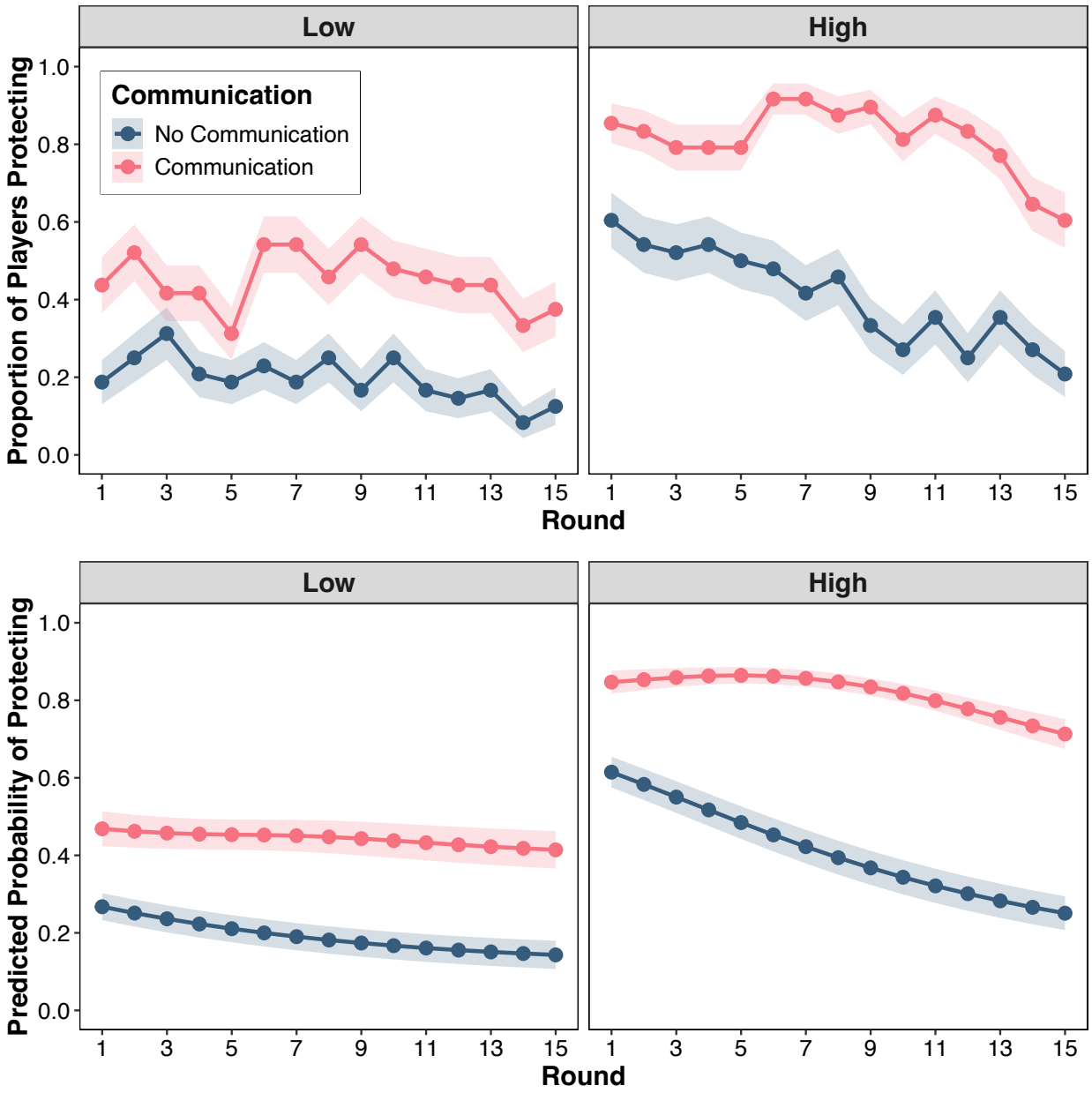


Figure 2

Observed proportion of players choosing to protect (top panels) and model estimates of the predicted probability of protecting (bottom panels), across the fifteen rounds, with and without communication, in the low-risk (left panels) and high-risk (right panels) treatments. The shaded ribbon around each line represents the standard error of the mean (top panels) and the predicted probabilities (bottom panels).

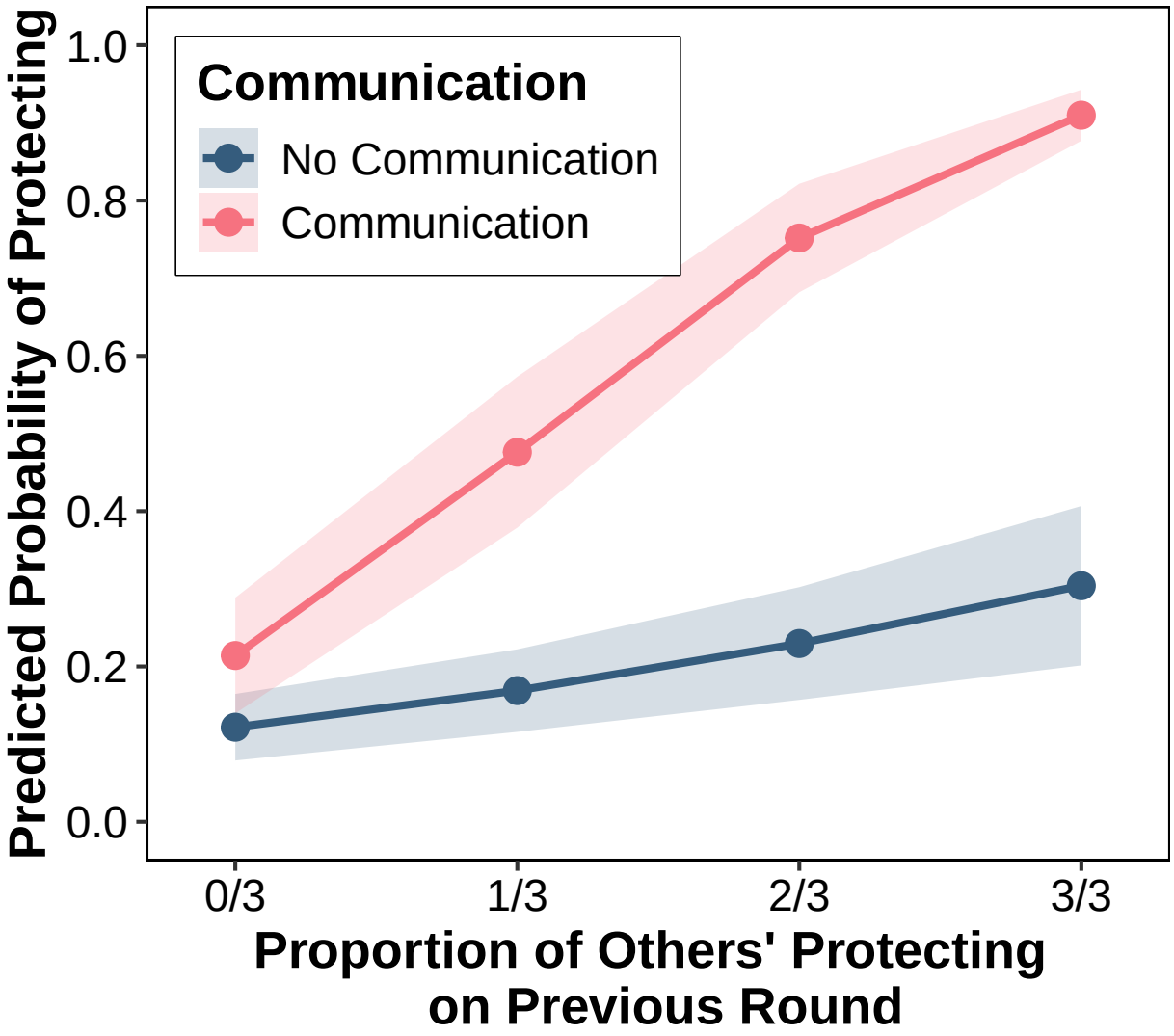


Figure 3

Predicted probability of protecting as a function of the proportion of other players protecting on the previous round, with and without communication. The shaded ribbon around each line represents the standard error of the predicted probabilities.

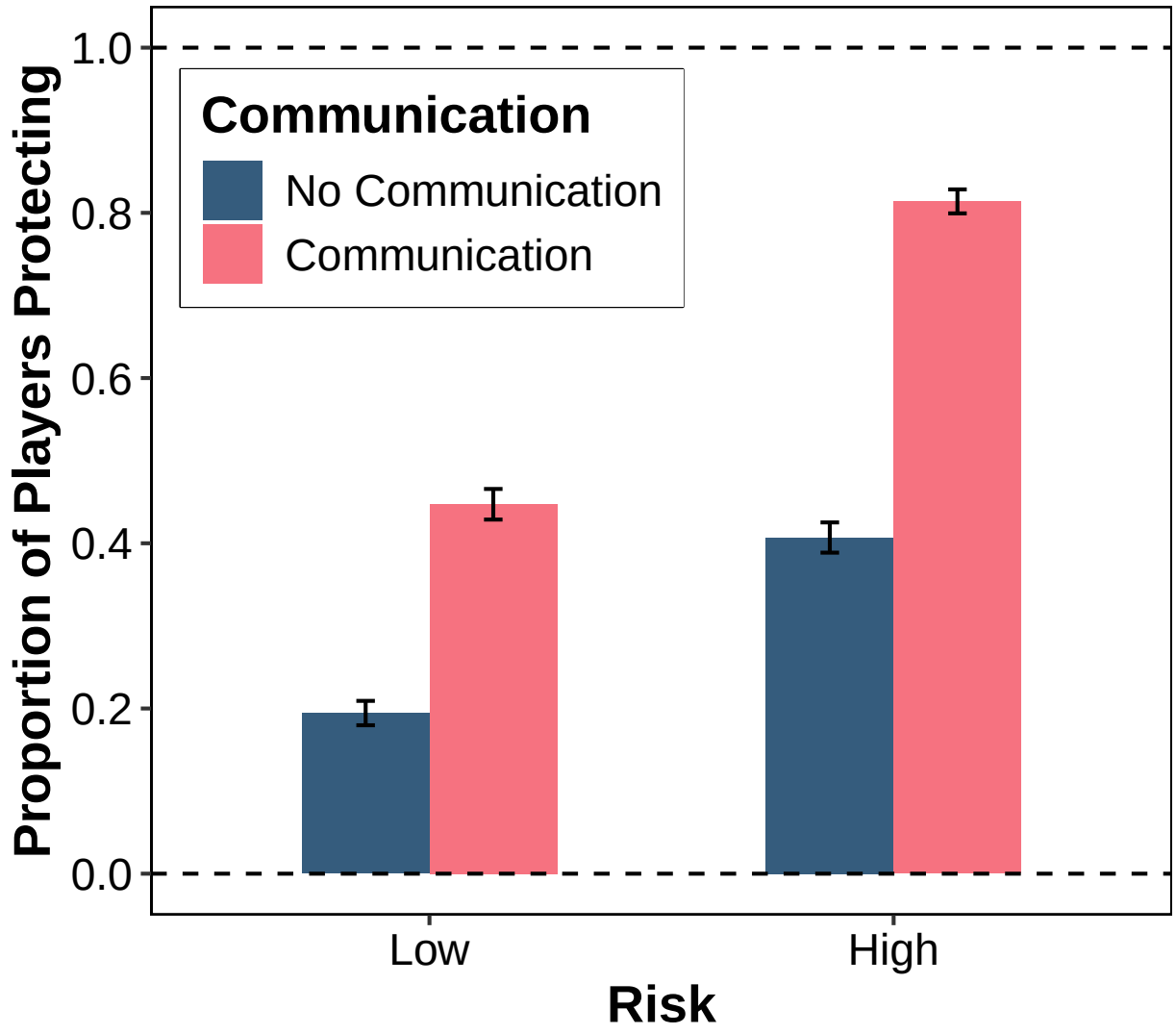


Figure 4

Proportion of players choosing to protect (without respect to round), with and without communication, in the low-risk and high-risk treatments. Error bars represent the standard error of the mean. Hatched reference lines at 0 and 1 indicate the Nash equilibrium and cooperative outcome, respectively.

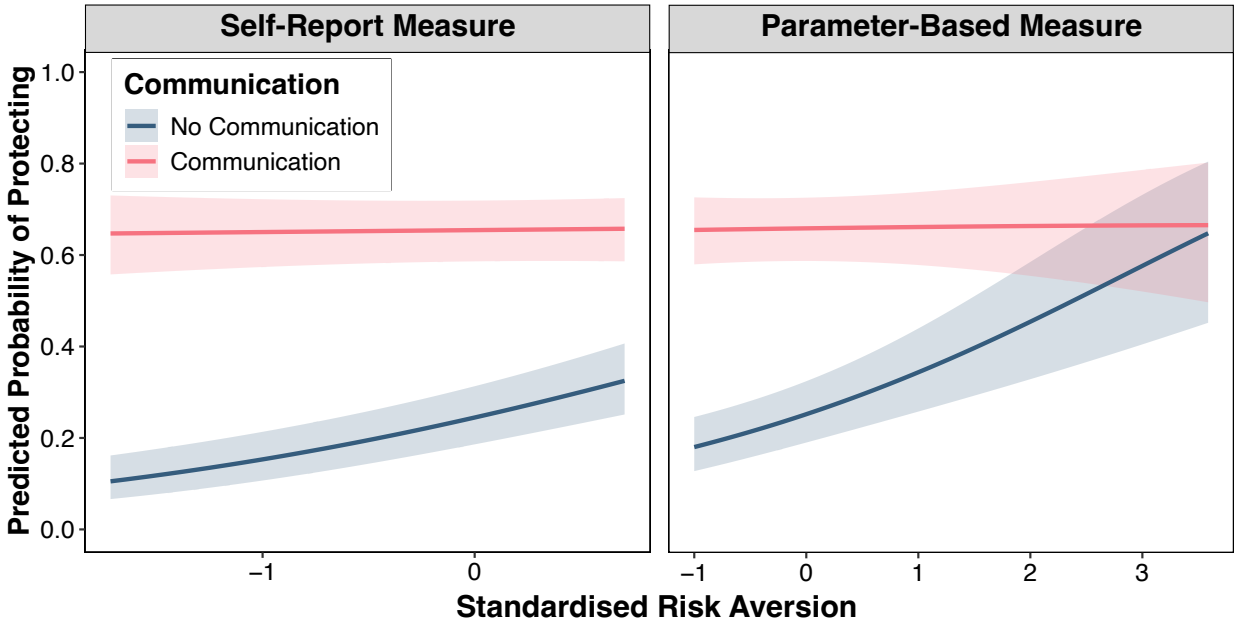


Figure 5

Predicted probability of protecting, with and without communication, as a function of standardised risk aversion based on the model using the self-report measure (left panel) and the model using the parameter-based measure (right panel). The shaded ribbon around each line represents the standard error of the predicted probabilities.

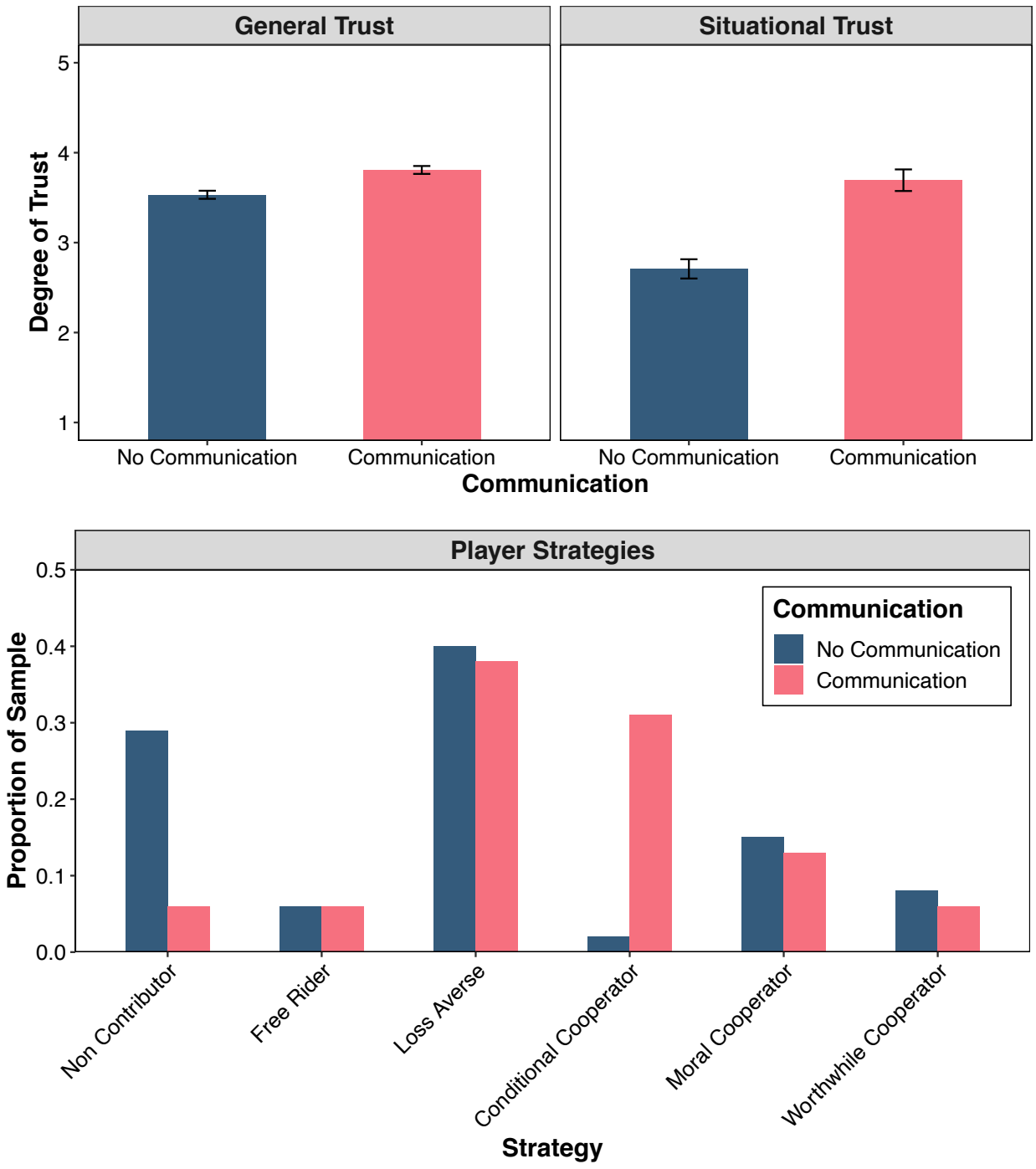


Figure 6

General and situational trust ratings (top left and top right panels, respectively) and player strategies (bottom panel) as a function of communication.



Figure 7

Word clouds of the communication text corpus. The left cloud shows the 100 most frequent words, ignoring sentiment, whereas the right cloud shows the 100 most frequent positive and negative words.

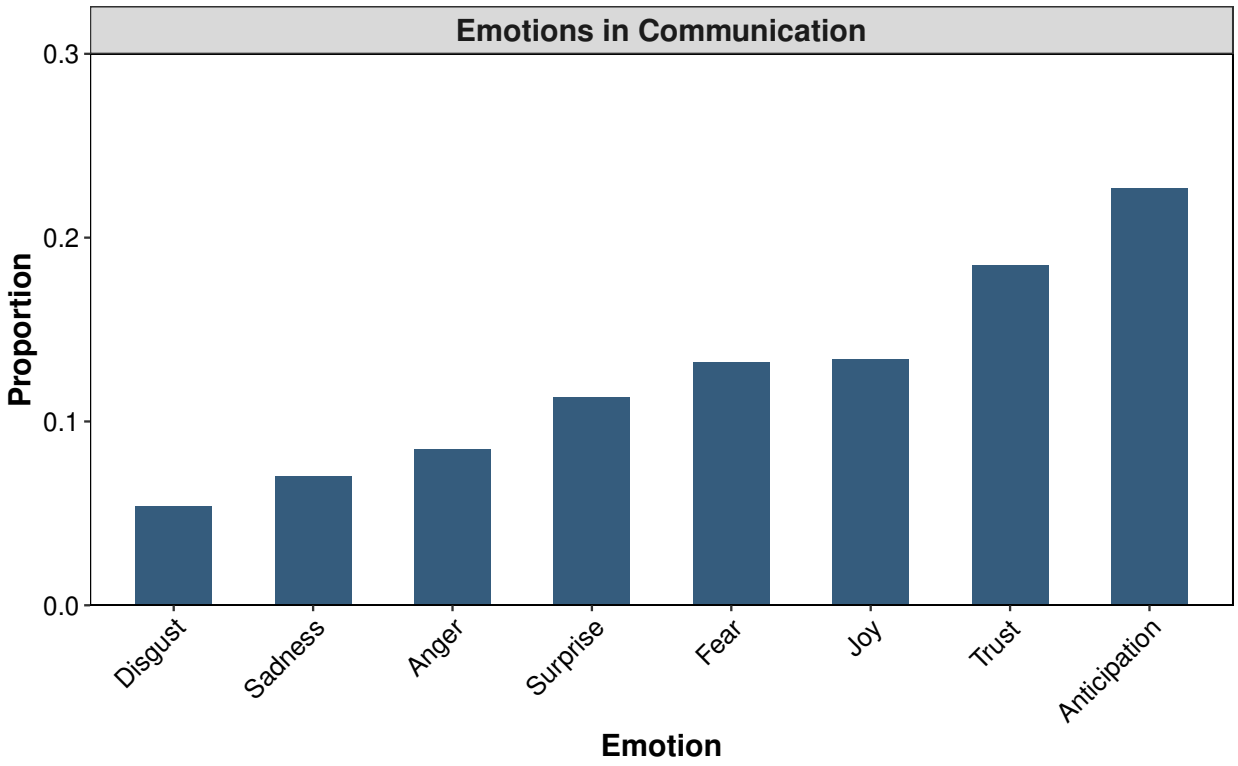


Figure 8

Proportion of words in the communication text corpus associated with each emotion.